

E2806544_1

The equation $x^5 + xy + y^3 + 4 = 0$ has no integer solutions

Size $h = 48$, length $l = 12$

Theorem 1. *The equation*

$$x^5 + xy + y^3 + 4 = 0 \tag{1}$$

has no integer solutions.

The proof uses 41 auxiliary polynomials $H_0, \dots, H_{40}$, 6 of which are constants, and a graph on these vertices with 140 edges; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) *Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.*
- (iii) *Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.*
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^5 + xy + y^3 + 4$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= -x - y, \\ H_2 &= -y - 1, \\ H_3 &= -y + 1, \\ H_4 &= x - y, \\ H_5 &= -2x - y, \\ H_6 &= -x - y - 1, \\ H_7 &= -x - y + 1, \\ H_8 &= -y - 2, \\ H_9 &= -y + 2, \\ H_{10} &= x - y - 1, \\ H_{11} &= 2x - y, \\ H_{12} &= -2x - y - 1, \\ H_{13} &= -2x - y + 1, \\ H_{14} &= -x - y - 2, \\ H_{15} &= -x - y + 2, \\ H_{16} &= x - y - 2, \\ H_{17} &= x^2 - y, \\ H_{18} &= x^2 - x + y, \\ H_{19} &= x^2 + y - 1, \\ H_{20} &= x^2 + y + 1, \\ H_{21} &= x^2 + x + y, \\ H_{22} &= x^2 - x - y, \\ H_{23} &= x^2 - y - 1, \\ H_{24} &= x^2 - y + 1, \\ H_{25} &= x^2 + x - y, \\ H_{26} &= x^2 - x + y + 1, \\ H_{27} &= x^2 + y - 2, \\ H_{28} &= x^2 + y + 2, \\ H_{29} &= x^2 + x + y - 1, \\ H_{30} &= x^2 + x + y + 1, \\ H_{31} &= x^2 + 2x + y, \\ H_{32} &= x^2 - y - 2, \\ H_{33} &= x^2 - y + 2, \\ H_{34} &= x^2 + x - y + 1, \\ H_{35} &= 1, \\ H_{36} &= -1, \\ H_{37} &= 2, \end{aligned}$$

$$\begin{aligned}
H_{38} &= 6, \\
H_{39} &= -3, \\
H_{40} &= 3.
\end{aligned}$$

The edges are $\{0, 1\}, \{0, 6\}, \{0, 14\}, \{0, 15\}, \{0, 18\}, \{0, 23\}, \{0, 26\}, \{0, 27\}, \{0, 33\}, \{1, 8\}, \{1, 12\}, \{1, 24\}, \{1, 27\}, \{1, 32\}, \{1, 33\}, \{2, 9\}, \{2, 10\}, \{2, 14\}, \{2, 16\}, \{2, 17\}, \{2, 18\}, \{2, 19\}, \{2, 21\}, \{2, 24\}, \{2, 25\}, \{2, 28\}, \{2, 34\}, \{3, 5\}, \{3, 18\}, \{3, 21\}, \{3, 23\}, \{3, 25\}, \{3, 26\}, \{3, 28\}, \{3, 30\}, \{4, 8\}, \{4, 21\}, \{4, 24\}, \{4, 32\}, \{5, 6\}, \{5, 10\}, \{5, 15\}, \{5, 26\}, \{5, 27\}, \{5, 28\}, \{5, 29\}, \{5, 34\}, \{6, 8\}, \{6, 10\}, \{6, 18\}, \{6, 19\}, \{6, 27\}, \{6, 28\}, \{6, 33\}, \{6, 34\}, \{7, 16\}, \{7, 22\}, \{7, 23\}, \{8, 14\}, \{8, 17\}, \{8, 20\}, \{8, 25\}, \{8, 27\}, \{8, 31\}, \{8, 33\}, \{9, 12\}, \{9, 14\}, \{9, 15\}, \{9, 17\}, \{9, 19\}, \{9, 24\}, \{9, 28\}, \{9, 31\}, \{10, 17\}, \{10, 20\}, \{10, 30\}, \{10, 34\}, \{11, 12\}, \{11, 16\}, \{11, 17\}, \{11, 25\}, \{11, 26\}, \{12, 26\}, \{12, 30\}, \{12, 34\}, \{13, 15\}, \{13, 21\}, \{14, 19\}, \{14, 20\}, \{14, 24\}, \{14, 26\}, \{14, 28\}, \{15, 16\}, \{15, 17\}, \{15, 18\}, \{15, 23\}, \{15, 26\}, \{15, 27\}, \{15, 31\}, \{15, 32\}, \{15, 33\}, \{17, 27\}, \{17, 33\}, \{18, 20\}, \{18, 22\}, \{18, 26\}, \{18, 34\}, \{20, 21\}, \{21, 25\}, \{21, 28\}, \{21, 32\}, \{21, 33\}, \{22, 27\}, \{23, 28\}, \{23, 29\}, \{23, 34\}, \{25, 26\}, \{25, 28\}, \{25, 29\}, \{26, 29\}, \{26, 34\}, \{27, 33\}, \{28, 30\}, \{30, 31\}, \{1, 36\}, \{5, 37\}, \{6, 38\}, \{8, 36\}, \{9, 37\}, \{10, 37\}, \{11, 39\}, \{13, 36\}, \{15, 37\}, \{16, 37\}, \{19, 39\}, \{20, 40\}, \{24, 36\}, \{27, 36\}, \{31, 39\}, \{32, 36\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^{10} - 216x^5 - 4x^3 - 432.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(4 + t)$ and the $\varphi_i(4 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-4 - t)$ and the $\varphi_i(-4 - t)$. Hence, for $x \geq 4$ and for $x \leq -4$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-3 \leq x \leq 3$. Checking the residues of y , we find

that for $x \in \{-3, -1, 1, 3\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , for $x = -2$ and $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , and for $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y . Hence (1) has no solution with $-3 \leq x \leq 3$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{35, 37, 38, 40\}$ and $H_i < 0$ for $i \in \{36, 39\}$. The other signs are given in Table 3 for $x \geq 4$ and for $x \leq -4$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 17\}$ be the set of primes dividing 2, the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p = (-1)^N$, where $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges, then $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so B_p is determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of B_p ; every integer solution lies in one of them. The result is $B_2 = -1, B_3 = 1, B_5 = 1, B_7 = 1, B_{11} = 1, B_{13} = 1, B_{17} = 1$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v = 1$, the product over all primes and $v = \infty$. By Step 3, the factors with $p \notin S$ are 1, so $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 4, $\prod_{p \in S} B_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$i = 0, m = 1$, neighbours 1, 6, 14, 15, 18, 23, 26, 27, 33	
W	$5x^4 - 4x^2 + 4x$
A	$x^9 + 4x^8y - x^8 + 6x^7y^2 - 4x^7y - 5x^7 + 4x^6y^3 - 5x^6y^2 - 9x^6y$ $+ 6x^6 + x^5y^4 - x^5y^3 - 3x^5y^2 + 15x^5y - x^5 + 2x^4y^4 - 4x^4y^3$ $+ 11x^4y^2 - 28x^4y - 9x^4 + x^3y^5 - 14x^3y^4 + 10x^3y^3 - 41x^3y^2$ $- 15x^3y + 4x^3 - 14x^2y^5 + 15x^2y^4 - 10x^2y^3 - 18x^2y^2 + 84x^2y$ $- 6xy^6 + 7xy^5 + 6xy^4 - 32xy^3 + 54xy^2 - 7xy - y^7 - 3y^6 + 2y^5 - 23y^4$ $- 11y^3 + 21y^2 - 51y$
B	$4x^4y^6 - 8x^4y^5 + 3x^4y^4 + 5x^4y^3 - 90x^4y^2 - 10x^4y - 128x^4$ $+ x^3y^7 - 18x^3y^6 + 6x^3y^5 - 7x^3y^4 - 114x^3y^3 - 8x^3y^2$ $- 56x^3y - 20x^3 - 15x^2y^7 + 11x^2y^6 + 3x^2y^5 - 81x^2y^4$ $+ 104x^2y^3 + 34x^2y^2 - 67x^2y + 320x^2 - 6xy^8 + 4xy^7 + 5xy^6 - 44xy^5$ $+ 57xy^4 - 10xy^3 - 115xy^2 + 165xy - 28x - y^9 - 4y^8 + 3y^7 - 18y^6 - 28y^5 + y^4$ $- 143y^3 - 12y^2 + 100y - 204$
$i = 1, m = 1$, neighbours 36, 0, 8, 12, 24, 27, 32, 33	
W	$x^4 + 2x^3 + 2x^2 + 4x$
A	$2x^4y^2 + 4x^4y + x^3y^3 + 3x^3y^2 + 2x^3y - x^3 - 4x^2y^3$ $- 10x^2y^2 - 4x^2y - 4x^2 - 2xy^4 - 7xy^3 - 7xy^2 - 2xy - 8x + y^4 + 5y^3 - 10y^2$ $- 4y$
B	$x^4y^4 - 3x^4y^3 - 14x^4y^2 + 20x^4y + 16x^4 + x^3y^5 + 3x^3y^4$ $+ 8x^3y^3 - 2x^3y^2 + 11x^3y + 20x^3 - 4x^2y^5 - 4x^2y^4 + 15x^2y^3$ $- 27x^2y^2 - 32x^2y + 12x^2 - 2xy^6 - 7xy^5 - 13xy^4 - 2xy^3 - 30xy^2 - 44xy$ $+ 2y^6 + 5y^5 - 19y^4 + 44y^2 - 40y - 32$
$i = 2, m = 1$, neighbours 9, 10, 14, 16, 17, 18, 19, 21, 24, 25, 28, 34	
W	$-6x^4 - 9x^3 - 18x^2 - 18x - 3$
A	$x^{14}y - 2x^{14} - x^{13}y^2 + 3x^{13}y - 2x^{13} - x^{12}y^3 - 4x^{12}y^2 + 9x^{12}y$ $+ 6x^{12} + x^{11}y^4 + x^{11}y^3 - 7x^{11}y^2 - 2x^{11}y + 8x^{11} + 2x^{10}y^4$ $- 3x^{10}y^3 + x^{10}y^2 - 14x^{10}y + 16x^{10} + 4x^9y^4 - 10x^9y^3$ $+ 11x^9y^2 - 22x^9y + 16x^9 + 7x^8y^5 + 16x^8y^4 - 37x^8y^3$ $- 11x^8y^2 - 60x^8y - 20x^8 - 3x^7y^6 + 2x^7y^5 + 19x^7y^4 - 12x^7y^3$ $+ 14x^7y^2 - 38x^7y - 60x^7 - x^6y^7 - 7x^6y^6 + 16x^6y^5 + 11x^6y^4$ $- 41x^6y^3 + 20x^6y^2 + 97x^6y - 58x^6 - 2x^5y^7 - 6x^5y^6 + 23x^5y^5$ $+ 18x^5y^4 - 33x^5y^3 - 77x^5y^2 + 121x^5y - 54x^5 - 16x^4y^7$ $- 27x^4y^6 + 62x^4y^5 + 111x^4y^4 + 58x^4y^3 - 158x^4y^2 + 105x^4y$ $- 26x^4 - x^3y^8 - 12x^3y^7 - 10x^3y^6 - 20x^3y^5 + 23x^3y^4$ $+ 240x^3y^3 + 57x^3y^2 + 224x^3y + 72x^3 + 3x^2y^9 + 5x^2y^8$ $- 29x^2y^7 - 18x^2y^6 + 27x^2y^5 - 25x^2y^4 + 151x^2y^3 - 21x^2y^2$ $+ 104x^2y + 144x^2 + xy^{10} + 7xy^9 - 7xy^8 - 22xy^7 + 36xy^6 - 19xy^5 - 163xy^4$ $+ 192xy^3 - 31xy^2 - 358xy - 81x + 2y^{10} + 16y^9 + 15y^8 - 60y^7 - 79y^6 + 28y^5$ $- 233y^4 + 113y^3 + 307y^2 - 422y - 340$
B	$-12x^4y^9 - 13x^4y^8 + 32x^4y^7 + 4x^4y^6 - 4x^4y^5 + 79x^4y^4$ $+ 607x^4y^3 - 252x^4y^2 - 120x^4y + 580x^4 - x^3y^{10} - 6x^3y^9$ $- 7x^3y^8 - 62x^3y^7 + 67x^3y^6 + 206x^3y^5 - 330x^3y^4 + 733x^3y^3$ $+ 330x^3y^2 + 34x^3y + 990x^3 + 3x^2y^{11} + 3x^2y^{10} - 23x^2y^9$ $+ 7x^2y^8 + 6x^2y^7 - 94x^2y^6 + 170x^2y^5 - 262x^2y^4 + 454x^2y^3$ $+ 247x^2y^2 - 673x^2y + 1008x^2 + xy^{12} + 6xy^{11} - 8xy^{10} + 12xy^9 + 45xy^8$ $- 190xy^7 - 85xy^6 + 509xy^5 - 913xy^4 + 1010xy^3 - 1556xy^2 - 216x + 2y^{12} + 13y^{11}$ $- 44y^9 + 43y^8 + 22y^7 - 523y^6 + 352y^5 + 83y^4 - 1453y^3 + 1565y^2 - 337y - 1351$
$i = 3, m = 1$, neighbours 5, 18, 21, 23, 25, 26, 28, 30	
W	$6x^4 + 8x^3 + 5x^2 - x - 19$
A	$-2x^{10} - x^9y - 2x^9 - 7x^8y - 2x^8 - 3x^7y^2 - 9x^7y - 2x^7 - 6x^6y^2$ $- 3x^6y + 4x^6 + x^5y^3 - 11x^5y^2 - 2x^5y + 12x^5 + x^4y^4 + 7x^4y^3$ $+ 14x^4y^2 + 16x^4y + 10x^4 + 12x^3y^4 + 10x^3y^3 + 17x^3y^2 + 43x^3y$ $- 26x^3 + 3x^2y^5 + 19x^2y^4 + 34x^2y^3 + 35x^2y^2 + 39x^2y - 86x^2$ $+ 11xy^5 + 24xy^4 + 42xy^3 + 55xy^2 + 9xy - 138x - y^6 + 13y^5 + 11y^4 - 7y^3$ $+ 36y^2 - 9y - 120$
B	$x^4y^6 + 9x^4y^5 + 26x^4y^4 + 21x^4y^3 + 38x^4y^2 + 129x^4y + 175x^4$ $+ 12x^3y^6 + 27x^3y^5 + 68x^3y^4 + 193x^3y^3 + 234x^3y^2 + 334x^3y$ $+ 418x^3 + 3x^2y^7 + 25x^2y^6 + 78x^2y^5 + 148x^2y^4 + 286x^2y^3$ $+ 372x^2y^2 + 515x^2y + 533x^2 + 11xy^7 + 35xy^6 + 95xy^5 + 214xy^4 + 319xy^3$ $+ 387xy^2 + 598xy + 514x - y^8 + 13y^7 + 29y^6 + 27y^5 + 122y^4 + 159y^3 + 11y^2$

$+155y + 119$
 $i = 4, m = 1, \text{ neighbours } 8, 21, 24, 32$
 $W \quad -x^4 + x^3 + 2x^2$
 $A \quad -x^3 + 2x^2 - xy + x + y$
 $B \quad -2x^4y - 6x^4 - x^3y^2 - 2x^3y - 2x^3 + x^2y + 6x^2 - xy^3 + 4xy^2 + 7xy - 4x$
 $\quad + 2y^3 + 3y^2$
 $i = 5, m = 1, \text{ neighbours } 37, 3, 6, 10, 15, 26, 27, 28, 29, 34$
 $W \quad -3x^4 + 36x^3 - 32x^2 + 24x - 20$
 $A \quad -2x^8y + 2x^8 - 2x^7y^2 + 4x^7y - 2x^7 + 2x^6y^3 - 6x^6y^2 + 10x^6y$
 $\quad - 6x^6 + 2x^5y^4 - 6x^5y^3 + 6x^5y^2 - 6x^5y + 4x^5 + 8x^4y^4$
 $\quad - 8x^4y^3 + 10x^4y^2 - 2x^4y - 8x^4 + 6x^3y^5 - 14x^3y^3 + 26x^3y^2$
 $\quad - 18x^3y - 9x^3 + 14x^2y^5 - 10x^2y^4 - 6x^2y^3 - 16x^2y^2 - 12x^2y$
 $\quad + 246x^2 + 2xy^6 + 22xy^5 - 52xy^4 + 38xy^3 + 12xy^2 - 32xy - 1478x - 2y^7 + 14y^6$
 $\quad - 14y^5 + 10y^4 - 78y^3 - 222y^2 - 858y - 96$
 $B \quad 2x^4y^5 + 10x^4y^4 - 32x^4y^3 - 128x^4y^2 - 432x^4y - 1267x^4$
 $\quad + 6x^3y^7 - 18x^3y^6 + 18x^3y^5 + 40x^3y^4 + 32x^3y^3 + 273x^3y^2$
 $\quad + 582x^3y + 1422x^3 + 8x^2y^7 - 4x^2y^6 - 2x^2y^5 - 24x^2y^4$
 $\quad - 176x^2y^3 - 286x^2y^2 - 586x^2y - 1510x^2 + 6xy^8 - 12xy^7 + 4xy^6$
 $\quad + 50xy^5 + 62xy^4 + 214xy^3 + 262xy^2 + 4xy + 1424x - 2y^9 + 16y^8 - 18y^7 - 12y^6$
 $\quad + 6y^5 - 248y^4 - 878y^3 - 434y^2 - 836y - 3424$
 $i = 6, m = 1, \text{ neighbours } 38, 0, 5, 8, 10, 18, 19, 27, 28, 33, 34$
 $W \quad 12x^4 - 24x^3 + 6x^2 + 42x - 36$
 $A \quad -12x^9y^2 - 24x^9y + 6x^8y^3 + 24x^8y^2 + 24x^8y + 6x^7y^4 - 6x^7y^3$
 $\quad - 48x^7y^2 - 24x^7y + 12x^6y^4 + 30x^6y^3 + 36x^6y^2 + 48x^6y$
 $\quad + 12x^5y^5 + 66x^5y^4 + 48x^5y^3 - 12x^5y^2 + 120x^5y + 18x^4y^5$
 $\quad + 12x^4y^4 - 30x^4y^3 + 24x^4y^2 - 24x^4y - 12x^3y^6 + 60x^3y^5$
 $\quad + 138x^3y^4 - 78x^3y^3 - 48x^3y^2 - 24x^3y - 144x^3 - 6x^2y^7$
 $\quad + 6x^2y^6 + 60x^2y^5 - 30x^2y^4 + 30x^2y^3 + 300x^2y^2 - 144x^2y$
 $\quad + 576x^2 - 36xy^7 - 54xy^6 + 84xy^5 - 96xy^4 - 444xy^3 - 240xy^2 - 240xy - 720x$
 $\quad - 12y^8 - 42y^7 + 6y^6 - 132y^5 - 624y^4 + 36y^3 + 828y^2 - 1104y - 1872$
 $B \quad 30x^4y^7 - 72x^4y^5 - 72x^4y^4 + 12x^4y^3 + 876x^4y^2 - 720x^4y$
 $\quad - 1152x^4 - 6x^3y^8 + 48x^3y^7 + 12x^3y^6 - 6x^3y^5 + 390x^3y^4$
 $\quad - 600x^3y^3 - 276x^3y^2 + 1536x^3y - 1692x^3 - 6x^2y^9 + 24x^2y^8$
 $\quad + 42x^2y^7 - 84x^2y^6 + 12x^2y^5 + 468x^2y^4 + 504x^2y^3$
 $\quad - 972x^2y^2 + 444x^2y + 3348x^2 - 24xy^9 - 12xy^7 - 90xy^6 + 12xy^5$
 $\quad - 1080xy^4 + 588xy^3 + 1200xy^2 - 4896xy + 288x - 12y^{10} - 36y^9 + 48y^8 - 174y^7$
 $\quad - 672y^6 + 588y^5 - 144y^4 - 3360y^3 + 1536y^2 + 1776y - 6192$
 $i = 7, m = 1, \text{ neighbours } 16, 22, 23$
 $W \quad -x^4 + 2x^2 - 3x + 2$
 $A \quad -x^3 + 4x + y - 3$
 $B \quad x^4y + 2x^4 - x^3y^2 - x^3y + 5x^3 - 5x^2y - 12x^2 + 3xy^2 + 5xy + 2x + y^3$
 $\quad - y^2 + 2y + 8$
 $i = 8, m = 1, \text{ neighbours } 36, 1, 4, 6, 14, 17, 20, 25, 27, 31, 33$
 $W \quad 2x^4 + 14x^2 + 24x - 40$
 $A \quad x^{11} + 2x^{10}y + 6x^{10} + 9x^9y + 14x^9 - 2x^8y^3 - 3x^8y^2 + 14x^8y + 18x^8$
 $\quad - x^7y^4 - 9x^7y^3 - 16x^7y^2 + 14x^7y + 13x^7 - 3x^6y^4 - 22x^6y^3$
 $\quad - 29x^6y^2 + 14x^6y - 16x^6 - 3x^5y^4 - 39x^5y^3 - 29x^5y^2 - 12x^5y$
 $\quad - 76x^5 + 7x^4y^5 - 4x^4y^4 - 51x^4y^3 - 6x^4y^2 - 52x^4y - 128x^4$
 $\quad + 5x^3y^6 + 22x^3y^5 + 3x^3y^4 - 24x^3y^3 + 72x^3y^2 - 49x^3y$
 $\quad - 144x^3 + x^2y^7 + 15x^2y^6 + 47x^2y^5 + 16x^2y^4 + 30x^2y^3$
 $\quad + 172x^2y^2 - 8x^2y - 100x^2 + 3xy^7 + 24xy^6 + 61xy^5 + xy^4 + 77xy^3$
 $\quad + 252xy^2 + 76xy - 8x - 5y^7 + 35y^6 + 59y^5 - 5y^4 + 172y^3 + 280y^2 + 232y + 208$
 $B \quad 10x^4y^7 - 10x^4y^6 - 23x^4y^5 + 30x^4y^4 - 85x^4y^3 - 22x^4y^2$
 $\quad - 65x^4y - 238x^4 + 5x^3y^8 + 16x^3y^7 - 5x^3y^6 + 20x^3y^5$
 $\quad + 86x^3y^4 - 109x^3y^3 + 230x^3y^2 - 172x^3y + 48x^3 + x^2y^9$
 $\quad + 13x^2y^8 + 25x^2y^7 - x^2y^6 + 135x^2y^5 + 77x^2y^4 + 11x^2y^3$
 $\quad + 274x^2y^2 + 216x^2y - 472x^2 + 3xy^9 + 15xy^8 + 29xy^7 + 6xy^6 + 214xy^5$
 $\quad + 39xy^4 + 166xy^3 + 248xy^2 + 744xy - 976x - 6y^9 + 47y^8 - 28y^7 + 33y^6 + 234y^5$
 $\quad + 40y^4 + 132y^3 + 632y^2 - 144y + 1216$
 $i = 9, m = 1, \text{ neighbours } 37, 2, 12, 14, 15, 17, 19, 24, 28, 31$
 $W \quad -24x^4 + 12x^3 + 66x^2 + 120x + 216$

$A \quad 4x^8y + 4x^8 + 10x^7y^2 + 20x^7y + 10x^7 + 8x^6y^3 + 36x^6y^2 + 24x^6y$
 $-4x^6 + 2x^5y^4 + 30x^5y^3 + 50x^5y^2 + 2x^5y - 20x^5 + 12x^4y^4$
 $+28x^4y^3 + 36x^4y^2 - 24x^4y - 44x^4 + 2x^3y^5 - 16x^3y^4 - 6x^3y^2$
 $-130x^3y - 682x^3 - 26x^2y^5 - 76x^2y^4 - 54x^2y^3 - 116x^2y^2$
 $-180x^2y + 508x^2 - 12xy^6 - 94xy^5 - 128xy^4 - 44xy^3 - 88xy^2 - 54xy + 3060x$
 $-2y^7 - 54y^6 - 114y^5 - 38y^4 + 24y^3 - 108y^2 + 12y + 4296$

$B \quad 16x^4y^6 + 64x^4y^5 + 170x^4y^4 + 300x^4y^3 + 544x^4y^2 + 1208x^4y$
 $+1694x^4 + 2x^3y^7 - 20x^3y^6 - 78x^3y^5 - 230x^3y^4 - 708x^3y^3$
 $-2198x^3y^2 - 4556x^3y - 9132x^3 - 28x^2y^7 - 170x^2y^6 - 512x^2y^5$
 $-1272x^2y^4 - 3002x^2y^3 - 5938x^2y^2 - 12438x^2y - 22472x^2 - 12xy^8$
 $-132xy^7 - 466xy^6 - 1078xy^5 - 2574xy^4 - 5786xy^3 - 8892xy^2 - 18076xy - 32040x$
 $-2y^9 - 60y^8 - 238y^7 - 510y^6 - 1188y^5 - 2958y^4 - 6092y^3 - 7784y^2 - 15984y$
 -31920

$i = 10, m = 1, \text{ neighbours } 37, 2, 5, 6, 17, 20, 30, 34$

$W \quad 2x^4 - 4x^3 + 14x^2 + 8x - 6$

$A \quad -4x^5y - 4x^5 - 6x^4y^2 - 18x^4y - 12x^4 - 2x^3y^3 - 16x^3y^2 - 38x^3y$
 $-28x^3 - 4x^2y^3 - 32x^2y^2 - 60x^2y - 16x^2 - 24xy^2 - 56xy - 104x + 28y^4$
 $+88y^3 + 88y^2 + 136y + 96$

$B \quad -12x^4y^4 - 52x^4y^3 - 76x^4y^2 - 152x^4y - 24x^4 - 2x^3y^5$
 $-22x^3y^4 - 52x^3y^3 - 136x^3y^2 - 88x^3y - 96x^3 - 4x^2y^5$
 $+4x^2y^4 - 46x^2y^3 - 48x^2y^2 - 10x^2y - 260x^2 + 36xy^5 + 44xy^4$
 $+12xy^3 + 164xy^2 + 84xy - 92x + 30y^6 + 64y^5 + 28y^4 + 216y^3 + 226y^2 + 124y + 420$

$i = 11, m = 1, \text{ neighbours } 39, 12, 16, 17, 25, 26$

$W \quad 12x^3 + 3x^2$

$A \quad 6x^3 - 3x^2y - 9x^2 - 3xy^2 - 15xy - 150x + 9y^2 + 75y$

$B \quad -3x^4y^2 - 27x^4y - 33x^4 - 15x^3y^2 - 30x^3y - 9x^3 - 12x^2y^3$
 $-24x^2y^2 - 9x^2y - 15x^2 - 3xy^4 + 6xy^3 + 24xy^2 + 81xy + 18x + 12y^4 + 87y^3$
 $+15y^2 + 42y + 300$

$i = 12, m = 1, \text{ neighbours } 1, 9, 11, 26, 30, 34$

$W \quad -8x^3 - 14x^2 - 3x$

$A \quad 2x^3y - 4x^3 + x^2y^2 - 4x^2 - xy^3 + 5xy^2 - 2xy - 72x + 2y^3 - 38y - 36$

$B \quad x^4y^3 - 3x^4y^2 - 14x^4y + 96x^4 + x^3y^3 + 4x^3y^2 - 38x^3y + 78x^3$
 $+2x^2y^4 - 9x^2y^3 + 18x^2y^2 - 13x^2y - 3x^2 - xy^5 + 6xy^4 - 7xy^3$
 $-5xy^2 - 26xy + y^5 - 35y^3 + 6y^2 - 8y - 144$

$i = 13, m = 1, \text{ neighbours } 36, 15, 21$

$W \quad -x^4 - 2x^3 + 5x^2 + 3$

$A \quad -x^3 - 4x^2 + 6x + 4y - 6$

$B \quad 2x^4y - 13x^4 - x^3y^2 + 7x^3y + 3x^3 - 4x^2y^2 - 7x^2 - 2xy^2 + 6xy + 4x$
 $+4y^3 - 2y^2 - 3y + 15$

$i = 14, m = 1, \text{ neighbours } 0, 2, 8, 9, 19, 20, 24, 26, 28$

$W \quad -2x^4 + 9x^3 - 5x^2 - 40x - 12$

$A \quad x^5y^4 + x^5y^3 - 4x^5y^2 - 4x^5y - x^4y^4 - x^4y^3 + 4x^4y^2$
 $+4x^4y + 3x^3y^5 + 7x^3y^4 - 8x^3y^3 - 28x^3y^2 - 16x^3y - 4x^3$
 $-2x^2y^5 - 5x^2y^4 + 5x^2y^3 + 20x^2y^2 + 12x^2y + 36x^2 + 2xy^6$
 $+11xy^5 + 5xy^4 - 40xy^3 - 52xy^2 - 16xy - 101x - y^7 - 3y^6 + y^5 - 4y^4 - 13y^3$
 $+108y^2 + 37y - 266$

$B \quad -2x^4y^6 + x^4y^5 + x^4y^4 - 24x^4y^3 + 72x^4y^2 + 17x^4y - 196x^4$
 $+3x^3y^7 + x^3y^6 - 16x^3y^5 + 22x^3y^4 + 10x^3y^3 - 129x^3y^2$
 $+166x^3y - 255x^3 - 2x^2y^7 + 3x^2y^6 + 2x^2y^5 - 32x^2y^4$
 $+70x^2y^3 - 12x^2y^2 - 93x^2y + 678x^2 + 3xy^8 + 3xy^7 - 10xy^6 + 15xy^5$
 $-27xy^4 - 79xy^3 + 169xy^2 - 565xy + 508x - y^9 + 4y^7 - 20y^6 - 3y^5 + 115y^4$
 $-182y^3 + 60y^2 + 300y - 460$

$i = 15, m = 1, \text{ neighbours } 37, 0, 5, 9, 13, 16, 17, 18, 23, 26, 27, 31, 32, 33$

$W \quad 90x^4 - 234x^3 - 324x^2 + 1080x - 864$

$A \quad 8x^{14}y^2 - 16x^{14}y - 20x^{13}y^2 + 40x^{13}y - 6x^{12}y^4 - 2x^{12}y^3$
 $-4x^{12}y^2 + 64x^{12}y - 2x^{11}y^5 + 2x^{11}y^4 + 8x^{11}y^3 + 116x^{11}y^2$
 $-248x^{11}y - 2x^{10}y^4 + 114x^{10}y^3 - 304x^{10}y^2 + 168x^{10}y + 2x^9y^5$
 $+68x^9y^4 - 370x^9y^3 + 332x^9y^2 + 240x^9y + 24x^8y^6 - 8x^8y^5$
 $-128x^8y^4 + 168x^8y^3 + 308x^8y^2 - 904x^8y + 14x^7y^7 - 16x^7y^6$
 $+26x^7y^5 - 50x^7y^4 + 498x^7y^3 - 1628x^7y^2 + 864x^7y + 2x^6y^8$
 $-2x^6y^7 + 2x^6y^6 - 200x^6y^5 + 446x^6y^4 - 1060x^6y^3$

$$\begin{aligned}
& +1416x^6y^2 + 848x^6y - 16x^5y^7 - 198x^5y^6 + 602x^5y^5 - 796x^5y^4 \\
& + 908x^5y^3 + 1368x^5y^2 - 2272x^5y - 38x^4y^8 - 38x^4y^7 + 516x^4y^6 \\
& - 632x^4y^5 + 484x^4y^4 + 348x^4y^3 - 2712x^4y^2 + 1056x^4y \\
& - 36x^3y^9 + 30x^3y^8 + 110x^3y^7 - 292x^3y^6 - 34x^3y^5 \\
& + 694x^3y^4 - 316x^3y^3 + 416x^3y^2 + 3104x^3y - 8100x^3 - 14x^2y^{10} \\
& + 14x^2y^9 - 54x^2y^8 + 46x^2y^7 - 290x^2y^6 + 830x^2y^5 \\
& + 404x^2y^4 - 656x^2y^3 + 3744x^2y^2 - 4544x^2y + 42120x^2 - 2xy^{11} \\
& + 2xy^{10} + 4xy^9 + 280xy^8 - 514xy^7 + 458xy^6 + 1100xy^5 - 3676xy^4 + 760xy^3 \\
& - 3232xy^2 - 2368xy + 3564x + 18y^{10} + 146y^9 - 466y^8 + 522y^7 + 156y^6 - 5088y^5 \\
& + 468y^4 - 2204y^3 + 7600y^2 + 72828y - 115992 \\
B \quad & -22x^4y^{10} - 76x^4y^9 + 288x^4y^8 - 210x^4y^7 - 668x^4y^6 \\
& + 340x^4y^5 - 3952x^4y^4 - 1988x^4y^3 + 12848x^4y^2 + 63996x^4y \\
& + 230040x^4 - 24x^3y^{11} - 22x^3y^{10} + 126x^3y^9 - 50x^3y^8 \\
& + 370x^3y^7 + 1322x^3y^6 - 322x^3y^5 - 4216x^3y^4 - 14824x^3y^3 \\
& - 45020x^3y^2 - 105924x^3y - 95904x^3 - 12x^2y^{12} - 10x^2y^{11} \\
& - 60x^2y^{10} - 186x^2y^9 - 170x^2y^8 + 258x^2y^7 + 1552x^2y^6 \\
& + 4432x^2y^5 + 12544x^2y^4 + 9752x^2y^3 + 12748x^2y^2 - 61408x^2y \\
& - 519696x^2 - 2xy^{13} - 2xy^{12} - 14xy^{11} + 82xy^{10} - 158xy^9 - 708xy^8 + 114xy^7 \\
& - 1404xy^6 - 548xy^5 + 12560xy^4 + 43020xy^3 + 99512xy^2 + 382268xy + 855360x \\
& + 20y^{12} + 184y^{11} - 120y^{10} + 372y^9 + 1568y^8 - 3864y^7 - 5300y^6 - 12148y^5 \\
& - 36984y^4 + 732y^3 - 123344y^2 - 216288y - 141264 \\
i = 16, m = 1, \text{ neighbours } 37, 2, 7, 11, 15 \\
W \quad & 2x^3 + 2x^2 - 2x + 4 \\
A \quad & -4x - 8 \\
B \quad & 4x^3 - 4x^2 - 6xy^2 + 6xy - 16x - 2y^3 + 2y - 8 \\
i = 17, m = 1, \text{ neighbours } 2, 8, 9, 10, 11, 15, 27, 33 \\
W \quad & -4x^5 - 6x^4 + 4x^3 - 8x^2 + 8x - 8 \\
A \quad & x^5y^2 - 6x^5y + 28x^5 - x^4y^3 + 7x^4y^2 - 50x^4y - 4x^4 - 4x^3 \\
& + 2x^2y^3 + x^2y^2 - 2x^2y - 64x^2 - xy^4 - 6xy^3 - 8xy^2 + 84xy + 48x + y^5 \\
& - 5y^4 + 53y^3 - 70y^2 - 16y - 16 \\
B \quad & -x^8y^2 + 6x^8y - 44x^8 + x^7y^3 - 7x^7y^2 + 50x^7y - 44x^7 - x^6y^3 \\
& + 6x^6y^2 - 44x^6y + x^5y^4 - 7x^5y^3 + 51x^5y^2 - 50x^5y + 40x^5 \\
& - x^4y^4 + 4x^4y^3 - 32x^4y^2 - 84x^4y - 72x^4 - 3x^3y^5 + 8x^3y^4 \\
& - 20x^3y^3 + 58x^3y^2 + 48x^3y - 16x^3 + 2x^2y^6 - 8x^2y^5 + 47x^2y^4 \\
& - 26x^2y^3 - 100x^2y^2 + 140x^2y - 208x^2 - 3xy^6 + 6xy^5 - 16xy^4 + 69xy^3 \\
& + 18xy^2 + 48xy + 240x + 2y^7 - 9y^6 + 50y^5 - 40y^4 - 44y^3 + 148y^2 - 264y - 32 \\
i = 18, m = 24, \text{ neighbours } 0, 2, 3, 6, 15, 20, 22, 26, 34 \\
W \quad & -192x^5 + 144x^4 - 96x^3 + 48x^2 - 96x - 288 \\
A \quad & 96x^8y^3 - 336x^8y^2 + 1944x^8y - 5844x^8 - 576x^5y^3 + 576x^5y \\
& - 98226x^5 + 96x^4y^5 - 528x^4y^4 - 1800x^4y^3 + 20124x^4y^2 \\
& - 90696x^4y - 172563x^4 - 576x^3y^5 + 1728x^3y^4 - 1728x^3y^2 + 576x^3y \\
& - 57600x^3 + 576x^2y^5 - 3456x^2y^4 + 576x^2y^3 + 3456x^2y^2 - 1152x^2y \\
& + 46080x^2 - 1152xy^5 + 6336xy^4 + 576xy^3 - 65682xy^2 - 588420xy + 262809x \\
& - 192y^7 + 1536y^6 - 3744y^5 - 4944y^4 + 80100y^3 + 276075y^2 - 276633y - 20736 \\
B \quad & -96x^{11}y^3 + 336x^{11}y^2 - 1944x^{11}y + 5844x^{11} - 96x^{10}y^3 \\
& + 336x^{10}y^2 - 1944x^{10}y + 5844x^{10} + 96x^9y^4 - 432x^9y^3 + 2280x^9y^2 \\
& - 7788x^9y + 5844x^9 + 192x^8y^4 - 768x^8y^3 + 4224x^8y^2 - 13632x^8y \\
& + 67206x^8 - 192x^7y^5 - 96x^7y^4 + 240x^7y^3 - 8904x^7y^2 + 75912x^7y \\
& + 295065x^7 - 96x^6y^6 - 48x^6y^5 - 1272x^6y^4 + 1476x^6y^3 \\
& + 6072x^6y^2 + 930x^6y + 318441x^6 + 96x^5y^6 + 48x^5y^5 - 1512x^5y^4 \\
& + 10380x^5y^3 - 69840x^5y^2 - 294135x^5y + 318441x^5 + 96x^4y^7 \\
& + 144x^4y^6 + 1320x^4y^5 - 2988x^4y^4 + 4884x^4y^3 - 11424x^4y^2 \\
& + 74070x^4y - 4272x^4 + 192x^3y^7 + 768x^3y^6 + 10560x^3y^5 \\
& - 13536x^3y^4 + 71574x^3y^3 + 101940x^3y^2 + 202521x^3y + 335640x^3 \\
& - 192x^2y^8 + 1728x^2y^7 + 1248x^2y^6 - 13296x^2y^5 + 77268x^2y^4 \\
& + 268521x^2y^3 - 55506x^2y^2 + 626601x^2y + 1088100x^2 + 192xy^8 - 1344xy^7 \\
& - 4704xy^6 + 12912xy^5 - 9858xy^4 + 214965xy^3 - 247659xy^2 + 243729xy + 1272420x \\
& + 192y^9 - 2112y^8 + 3744y^7 + 8592y^6 - 85092y^5 - 265131y^4 + 294105y^3 \\
& - 297936y^2 - 1103148y + 1106532 \\
i = 19, m = 1, \text{ neighbours } 39, 2, 6, 9, 14 \\
W \quad & -3x^2 + 3x + 9 \\
A \quad & -3x - 3y - 6
\end{aligned}$$

B	$3x^4 + 3x^3y + 6x^3 - 3x^2y - 6x^2 - 3xy^2 - 3xy + 24x + 9y + 45$
$i = 20, m = 1$, neighbours 40, 8, 10, 14, 18, 21	
W	$x^5 - 3x^4 + x^3 - 4x^2 - 3x + 3$
A	$-x^4y + 9x^4 - 11x^3 + 24x^2 - 6xy + y^3 - 6y^2 + 3y + 7$
B	$-x^8 + x^7y - 3x^7 + x^6y + x^6 - x^5y^2 + 2x^5y - 7x^5 - x^4y^2 + 7x^4y$ $-14x^4 + 6x^3y^2 - 4x^3y - 10x^3 + x^2y^4 - 11x^2y^3 - 15x^2y^2$ $+7x^2y - 32x^2 + 5xy^3 + 4xy^2 - xy + 18x - y^5 + 4y^4 - 22y^3 - 13y^2 + 25y - 37$
$i = 21, m = 8$, neighbours 2, 3, 4, 13, 20, 25, 28, 32, 33	
W	$80x^5 + 48x^4 - 80x^3 + 96x^2 + 64x - 128$
A	$-32x^8y^3 - 688x^8y^2 - 2296x^8y - 2892x^8 - 128x^7y^2 + 128x^7$ $+64x^6y^3 - 64x^6y^2 - 64x^6y + 64x^6 + 64x^5y^4 + 128x^5y^3$ $-384x^5y^2 - 128x^5y - 9742x^5 + 96x^4y^5 + 1936x^4y^4 + 4168x^4y^3$ $-1388x^4y^2 - 13928x^4y - 9153x^4 - 64x^3y^5 + 512x^3y^4 + 960x^3y^3$ $-64x^3y^2 - 896x^3y + 10048x^3 - 128x^2y^4 - 256x^2y^3 + 768x^2y^2$ $+256x^2y - 8320x^2 - 192xy^6 - 640xy^5 + 1824xy^3 + 13530xy^2 - 7104xy - 24063x$ $-64y^7 - 1088y^6 - 3712y^5 - 6016y^4 - 4516y^3 - 22335y^2 - 28159y - 4096$
B	$32x^{11}y^3 + 688x^{11}y^2 + 2296x^{11}y + 2892x^{11} - 32x^{10}y^3 - 688x^{10}y^2$ $-2296x^{10}y - 2892x^{10} - 32x^9y^4 - 656x^9y^3 - 1608x^9y^2 - 596x^9y$ $+2892x^9 + 64x^8y^4 + 1344x^8y^3 + 3904x^8y^2 + 3488x^8y + 770x^8$ $-64x^7y^5 - 1312x^7y^4 - 3152x^7y^3 + 184x^7y^2 + 10376x^7y + 895x^7$ $+32x^6y^6 + 688x^6y^5 + 2264x^6y^4 + 2268x^6y^3 - 920x^6y^2$ $-1962x^6y + 10673x^6 + 32x^5y^6 + 624x^5y^5 + 888x^5y^4 - 2452x^5y^3$ $-9456x^5y^2 + 1067x^5y - 10673x^5 - 32x^4y^7 - 720x^4y^6 - 2888x^4y^5$ $-3156x^4y^4 + 3340x^4y^3 - 256x^4y^2 + 4274x^4y + 7216x^4 + 960x^3y^6$ $+4224x^3y^5 + 7136x^3y^4 + 21442x^3y^3 + 37948x^3y^2 + 43967x^3y$ $+64776x^3 - 64x^2y^8 - 1088x^2y^7 - 2432x^2y^6 - 2304x^2y^5$ $-7364x^2y^4 - 27809x^2y^3 - 33150x^2y^2 - 4751x^2y - 15108x^2 + 128xy^8$ $-320xy^7 - 2624xy^6 - 4384xy^5 - 14206xy^4 - 23067xy^3 - 11201xy^2 - 49913xy$ $-58364x + 64y^9 + 1024y^8 + 3584y^7 + 6656y^6 + 9636y^5 + 36607y^4 + 51071y^3$ $+22416y^2 + 89852y + 112636$
$i = 22, m = 2$, neighbours 7, 18, 27	
W	$2x^4 - 4x^3 - 2x + 4$
A	$2x^4 - 4x^3 + 16x^2 - 10x - 2y^2 - 18y - 4$
B	$-2x^7 - 2x^6 - 2x^5y - 2x^5 - 4x^4y - 8x^4 + 10x^3y - 2x^2y^3 - 4x^2y^2$ $+2x^2y - 32x^2 + 2xy^3 + 6xy^2 - 18xy + 20x - 2y^4 - 18y^3 - 20y - 64$
$i = 23, m = 1$, neighbours 0, 3, 7, 15, 28, 29, 34	
W	$4x^4 + 9x^3 - x^2 - 3x + 6$
A	$-x^5y^2 + 5x^5y - 7x^5 + x^4y^3 - 8x^4y^2 + 22x^4y - 21x^4 + x^3y^2$ $-x^3y - 16x^3 + 2x^2y^3 - 2x^2y^2 - 4x^2y - 60x^2 + xy^4 + 2xy^2 - 11xy + 2x$ $-y^5 + 4y^4 + 7y^3 - 34y^2 + 49y + 57$
B	$x^8y^2 - 5x^8y + 7x^8 - x^7y^3 + 8x^7y^2 - 22x^7y + 21x^7 + x^6y^3$ $-4x^6y^2 + 2x^6y + 7x^6 - x^5y^4 + 7x^5y^3 - 15x^5y^2 + 4x^5y + 9x^5$ $+x^4y^4 - x^4y^3 - 12x^4y^2 + 28x^4y - 68x^4 + x^3y^5 - 5x^3y^3$ $+5x^3y^2 - 34x^3y + 22x^3 - x^2y^6 + 10x^2y^5 - 20x^2y^4 - 2x^2y^3$ $+46x^2y^2 - 107x^2y + 21x^2 + xy^6 + xy^4 - 4xy^3 - 9xy^2 + 47xy - 28x - y^7$ $+6y^6 - 3y^5 - 33y^4 + 110y^3 - 48y^2 - 72y + 264$
$i = 24, m = 1$, neighbours 36, 1, 2, 4, 9, 14	
W	$-x^4 - x^2 + 2$
A	$-x^3 + xy - x + y^2 - y - 2$
B	$-x^4y - x^4 - x^3y^2 + x^3y + 2x^3 - x^2y^2 + x^2y + 4x^2 + xy^2 + 2xy + 4x$ $-2y^3 + 8y + 4$
$i = 25, m = 72$, neighbours 2, 3, 8, 11, 21, 26, 28, 29	
W	$-6048x^5 - 5184x^4 - 3744x^3 - 3312x^2 + 2016x - 7488$
A	$4320x^8y^2 + 87408x^8y + 977448x^8 - 46745652x^5 + 4320x^4y^4$ $+16560x^4y^3 - 591000x^4y^2 - 20229504x^4y + 4333686x^4 + 5184x^3y^4$ $-25920x^3y^3 - 72140544x^3 - 36288x^2y^4 - 82944x^2y^3 + 15552x^2y^2$ $+82944x^2y - 78859008x^2 + 20736xy^5 + 15552xy^4 - 103680xy^3 - 10387668xy^2$ $+176119656xy + 53777874x - 8640y^6 - 9792y^5 + 212160y^4 + 22684440y^3$ $-60952854y^2 - 46229970y - 14017536$
B	$-4320x^{11}y^2 - 87408x^{11}y - 977448x^{11} + 4320x^{10}y^2 + 87408x^{10}y$ $+977448x^{10} - 4320x^9y^3 - 91728x^9y^2 - 1064856x^9y - 977448x^9$

$$\begin{aligned}
& +8640x^8y^3 + 179136x^8y^2 + 2042304x^8y + 11144796x^8 - 8640x^7y^4 \\
& -131616x^7y^3 - 761136x^7y^2 + 16242672x^7y - 78163410x^7 - 4320x^6y^5 \\
& -70128x^6y^4 - 666696x^6y^3 + 2786160x^6y^2 - 5447508x^6y + 74253618x^6 \\
& -4320x^5y^5 - 61488x^5y^4 - 94440x^5y^3 + 13456512x^5y^2 \\
& -72715902x^5y - 74253618x^5 - 4320x^4y^6 - 65808x^4y^5 - 605208x^4y^4 \\
& +2880600x^4y^3 - 8594112x^4y^2 + 17678460x^4y - 3432864x^4 + 8640x^3y^6 \\
& -36864x^3y^5 + 162816x^3y^4 + 35233308x^3y^3 + 17200536x^3y^2 \\
& -49994322x^3y + 109939056x^3 - 8640x^2y^7 - 59904x^2y^6 + 69312x^2y^5 \\
& +22800408x^2y^4 - 48419154x^2y^3 + 2919684x^2y^2 + 199785906x^2y \\
& -200762184x^2 + 12096xy^7 - 2880xy^6 + 88320xy^5 + 12515844xy^4 \\
& +137843178xy^3 - 52929558xy^2 - 11018946xy + 446649672x - 8640y^8 - 14976y^7 \\
& +191424y^6 + 22639512y^5 - 60950550y^4 - 45345042y^3 + 76699488y^2 - 243832152y \\
& -184919880
\end{aligned}$$

$i = 26$, $m = 48$, neighbours 0, 3, 5, 11, 12, 14, 15, 18, 25, 29, 34

$$\begin{aligned}
W & -144x^5 - 864x^4 + 1152x^3 - 1008x^2 + 1008x + 1296 \\
A & 384x^8y^5 - 3648x^8y^4 - 10080x^8y^3 + 73968x^8y^2 - 204888x^8y \\
& +466140x^8 + 46080x^7y^3 - 46080x^7y + 32256x^6y^4 + 78336x^6y^3 \\
& -165888x^6y^2 + 55296x^6y - 2304x^5y^5 + 73728x^5y^4 - 34560x^5y^3 \\
& -258048x^5y^2 + 221184x^5y + 3273714x^5 + 384x^4y^7 - 10560x^4y^6 \\
& -33888x^4y^5 + 110064x^4y^4 + 54312x^4y^3 - 1004004x^4y^2 \\
& +3858144x^4y + 5444625x^4 - 2304x^3y^7 - 18432x^3y^6 - 25344x^3y^5 \\
& +239616x^3y^4 - 423936x^3y^3 + 285696x^3y^2 - 55296x^3y - 414720x^3 \\
& -4608x^2y^7 - 115200x^2y^6 + 142848x^2y^5 + 64512x^2y^4 \\
& -714240x^2y^3 + 681984x^2y^2 - 55296x^2y + 1700352x^2 - 46080xy^7 \\
& -145152xy^6 + 335232xy^5 - 371520xy^4 - 417888xy^3 + 4623714xy^2 + 17805204xy \\
& +5826087x - 768y^9 + 7680y^8 - 53760y^7 - 170880y^6 + 307392y^5 + 125088y^4 \\
& -3330396y^3 - 13452441y^2 - 11229939y - 1376790 \\
B & -384x^{11}y^5 + 3648x^{11}y^4 + 10080x^{11}y^3 - 55536x^{11}y^2 + 186456x^{11}y \\
& -466140x^{11} - 384x^{10}y^5 + 3648x^{10}y^4 + 10080x^{10}y^3 - 55536x^{10}y^2 \\
& +186456x^{10}y - 466140x^{10} + 384x^9y^6 - 3648x^9y^5 - 10080x^9y^4 \\
& +55536x^9y^3 - 186456x^9y^2 + 466140x^9y + 768x^8y^6 - 6912x^8y^5 \\
& -23808x^8y^4 + 100992x^8y^3 - 317376x^8y^2 + 745824x^8y - 2828310x^8 \\
& -768x^7y^7 + 4992x^7y^6 + 44352x^7y^5 - 72480x^7y^4 - 20784x^7y^3 \\
& +519672x^7y^2 - 3915576x^7y - 8521767x^7 - 384x^6y^8 + 2112x^6y^7 \\
& +21216x^6y^6 + 18000x^6y^5 + 51384x^6y^4 - 230220x^6y^3 \\
& -130920x^6y^2 - 1087266x^6y - 7558017x^6 + 384x^5y^8 - 2112x^5y^7 \\
& -28128x^5y^6 + 46128x^5y^5 + 144648x^5y^4 - 729108x^5y^3 \\
& +3264984x^5y^2 + 11350077x^5y + 963750x^5 + 384x^4y^9 - 1344x^4y^8 \\
& -25440x^4y^7 - 67344x^4y^6 - 26712x^4y^5 + 356316x^4y^4 \\
& -277248x^4y^3 + 688896x^4y^2 - 1387686x^4y - 82944x^4 + 768x^3y^9 \\
& +8448x^3y^8 - 59904x^3y^7 + 80256x^3y^6 + 19392x^3y^5 - 123360x^3y^4 \\
& -1443702x^3y^3 - 3447132x^3y^2 - 8727207x^3y - 8327256x^3 - 768x^2y^{10} \\
& +16896x^2y^9 + 76032x^2y^8 - 124800x^2y^7 + 252096x^2y^6 \\
& +822432x^2y^5 - 4579164x^2y^4 - 7358247x^2y^3 + 1101786x^2y^2 \\
& -22237569x^2y - 31121820x^2 + 768xy^{10} + 30720xy^9 + 189696xy^8 - 370944xy^7 \\
& +608640xy^6 + 705216xy^5 - 3385350xy^4 - 836181xy^3 + 12114909xy^2 - 6662274xy \\
& -22089540x + 768y^{11} - 10752y^{10} + 69120y^9 + 114048y^8 - 475200y^7 + 567456y^6 \\
& +3464892y^5 + 8748765y^4 + 1980822y^3 + 12717552y^2 + 41092212y + 3827544
\end{aligned}$$

$i = 27$, $m = 1$, neighbours 36, 0, 1, 5, 6, 8, 15, 17, 22, 33

$$\begin{aligned}
W & -6x^5 + 2x^4 + 46x^3 - 68x^2 - 52x + 120 \\
A & x^5y^4 - x^5y^3 - 108x^5y^2 - 354x^5y - 150x^5 - x^4y^5 - 2x^4y^4 \\
& +102x^4y^3 + 658x^4y^2 + 1172x^4y + 366x^4 - 9x^3y^4 + 34x^3y^2 \\
& -4x^3y + 548x^3 - 5x^2y^5 + 23x^2y^4 + 46x^2y^3 + 64x^2y^2 + 358x^2y \\
& -2764x^2 - xy^6 + 32xy^5 + 10xy^4 - 196xy^3 - 306xy^2 + 1808xy - 704x + y^7 + y^6 \\
& -153y^5 - 106y^4 + 1428y^3 + 1096y^2 - 7472y + 4336 \\
B & -x^8y^4 + x^8y^3 + 106x^8y^2 + 350x^8y + 114x^8 + x^7y^5 + 2x^7y^4 \\
& -109x^7y^3 - 668x^7y^2 - 1164x^7y - 342x^7 + x^6y^5 - 3x^6y^4 \\
& -104x^6y^3 - 138x^6y^2 + 586x^6y + 228x^6 - x^5y^6 + 114x^5y^4 \\
& +449x^5y^3 - 278x^5y^2 - 2336x^5y + 1080x^5 - x^4y^6 + 3x^4y^5 \\
& +100x^4y^4 + 142x^4y^3 - 162x^4y^2 - 706x^4y - 1308x^4 - x^3y^7 \\
& +15x^3y^6 + 119x^3y^5 + 61x^3y^4 - 654x^3y^3 - 68x^3y^2 + 2754x^3y \\
& +6328x^3 + x^2y^8 + 6x^2y^7 - 143x^2y^6 - 749x^2y^5 - 978x^2y^4
\end{aligned}$$

$$\begin{aligned}
& -426x^2y^3 - 2294x^2y^2 - 5340x^2y - 4400x^2 + xy^8 - 25xy^7 - 66xy^6 \\
& + 189xy^5 + 682xy^4 - 1880xy^3 - 3368xy^2 + 1960xy - 7648x - y^9 - 2y^8 + 148y^7 \\
& + 392y^6 - 644y^5 - 1764y^4 + 4368y^3 - 1312y^2 - 7008y + 15872
\end{aligned}$$

$i = 28, m = 1$, neighbours 2, 3, 5, 6, 9, 14, 21, 23, 25, 30

W $-20x^5 - 36x^4 - 89x^3 - 68x^2 - 42x - 20$

A $2x^6y^3 - 4x^6y^2 - 2x^6y + 4x^6 + x^5y^5 + 23x^5y^4 - 82x^5y^3$
 $+ 190x^5y^2 - 914x^5y + 4119x^5 - x^4y^6 - 9x^4y^5 + 190x^4y^4$
 $- 663x^4y^3 + 1935x^4y^2 - 9035x^4y + 21091x^4 + x^3y^6 + 13x^3y^5$
 $+ 11x^3y^4 - 67x^3y^3 - 72x^3y^2 + 54x^3y - 4796x^3 + 3x^2y^6$
 $+ 5x^2y^5 - 28x^2y^4 + 50x^2y^3 - 224x^2y^2 + 940x^2y + 8317x^2$
 $- 12xy^6 + xy^5 - 44xy^4 + 322xy^3 - 875xy^2 - 17119xy - 26264x + y^8 + 16y^7$
 $- 141y^6 - 53y^5 + 9y^4 + 3469y^3 + 7200y^2 - 37319y - 66610$

B $-x^8y^5 - 18x^8y^4 + 84x^8y^3 - 219x^8y^2 + 912x^8y - 4495x^8$
 $+ x^7y^6 + 13x^7y^5 - 174x^7y^4 + 639x^7y^3 - 2007x^7y^2 + 9055x^7y$
 $- 22475x^7 + x^6y^6 + 20x^6y^5 - 48x^6y^4 + 51x^6y^3 - 474x^6y^2$
 $+ 2671x^6y + 8990x^6 - x^5y^7 - 15x^5y^6 + 149x^5y^5 - 273x^5y^4$
 $+ 645x^5y^3 - 4822x^5y^2 + 3453x^5y + 27517x^5 - x^4y^7 - 24x^4y^6$
 $- 28x^4y^5 + 213x^4y^4 - 66x^4y^3 + 101x^4y^2 - 1394x^4y - 6277x^4$
 $- x^3y^8 - 30x^3y^7 + 79x^3y^6 - 346x^3y^5 + 1525x^3y^4$
 $- 5512x^3y^3 + 7216x^3y^2 - 14691x^3y - 20892x^3 + x^2y^9 + 8x^2y^8$
 $- 183x^2y^7 + 707x^2y^6 - 2011x^2y^5 + 8137x^2y^4 - 18252x^2y^3$
 $- 6696x^2y^2 + 49879x^2y - 85386x^2 + 16xy^8 - 38xy^7 + 245xy^6 - 765xy^5$
 $+ 2369xy^4 + 9123xy^3 - 436xy^2 + 41699xy + 51688x - y^{10} - 13y^9 + 170y^8 - 293y^7$
 $+ 499y^6 - 3914y^5 + 847y^4 + 35601y^3 - 18464y^2 + 8128y + 133020$

$i = 29, m = 1$, neighbours 5, 23, 25, 26

W $-x^4 + 3x^3 + 3x^2 - 4x - 1$

A $-x^3 + 4x^2 - xy + 5x + 4y - 5$

B $-8x^4 - 2x^3y - 11x^3 + x^2y^2 + 8x^2y + 14x^2 + xy^3 + xy^2 - 2xy + 9x - 5y^3$
 $- 2y^2 - 3y - 19$

$i = 30, m = 2$, neighbours 3, 10, 12, 28, 31

W $-2x^5 - 2x^3 + 2x^2 + 2x$

A $-4x^5 + 2x^4y - 5x^4 - 8x^3 + 8x^2 + 12xy + 9x - 2y^3 - y^2 + y$

B $-2x^7y + 5x^7 + 2x^6y - 5x^6 + 2x^5y^2 - 5x^5y - 4x^4y^2 + 8x^4y - 8x^4$
 $+ 4x^3y^3 + 2x^3y^2 + 17x^3y + 20x^3 - 2x^2y^4 + x^2y^3 - 2x^2y^2$
 $- 13x^2y + 8x^2 - 2xy^4 - 15xy^3 - 17xy^2 - 20x + 2y^5 - 5y^4 - 8y^3 + 12y^2 + 4y$

$i = 31, m = 2$, neighbours 39, 8, 9, 15, 30

W $-2x^5 - 8x^4 - 16x^3 - 14x^2 - 8x + 8$

A $-5x^5 - 49x^4 - 128x^3 - 312x^2 + 5xy^2 + 12xy - 428x + 33y^2 - 224y + 8$

B $x^8 + 15x^7 - x^6y - 30x^6 - 13x^5y + 60x^5 - 4x^4y^2 + 49x^4y - 204x^4$
 $+ 5x^3y^3 - 12x^3y^2 + 115x^3y - 124x^3 + 43x^2y^3 - 25x^2y^2 + 102x^2y$
 $+ 120x^2 - 5xy^4 + 54xy^3 - 53xy^2 + 232xy + 256x - 33y^4 + 236y^3 - 20y^2 - 204y + 944$

$i = 32, m = 1$, neighbours 36, 1, 4, 15, 21

W $-x^4 - x^3 + 5x^2 + 4x - 4$

A $-x^3 - 3x^2 + xy + 2x + 4y + 4$

B $x^5 - x^4y + 7x^4 - 3x^3y + 15x^3 - x^2y^2 + 5x^2y - 10x^2 + xy^3 - 2xy^2$
 $+ 10xy - 12x + 3y^3 + 16$

$i = 33, m = 1$, neighbours 0, 1, 6, 8, 15, 17, 21, 27

W $6x^4 - 2x^3 - 12x^2 - 28x - 24$

A $x^5y^3 + 13x^5y^2 + 54x^5y + 74x^5 - x^4y^4 - 16x^4y^3 - 94x^4y^2$
 $- 238x^4y - 222x^4 - 3x^3y^3 - 6x^3y^2 - 36x^3 - 3x^2y^4 - 9x^2y^3$
 $- 12x^2y^2 - 44x^2y + 68x^2 - xy^5 - 6xy^4 + 14xy^3 + 78xy^2 - 40xy - 320x + y^6$
 $+ 15y^5 + 45y^4 - 80y^3 - 382y^2 - 204y + 1264$

B $-x^8y^3 - 13x^8y^2 - 54x^8y - 74x^8 + x^7y^4 + 16x^7y^3 + 93x^7y^2$
 $+ 236x^7y + 222x^7 - x^6y^4 - 11x^6y^3 - 28x^6y^2 + 34x^6y + 148x^6$
 $+ x^5y^5 + 14x^5y^4 + 62x^5y^3 + 63x^5y^2 - 196x^5y - 488x^5 - x^4y^5$
 $- 11x^4y^4 - 32x^4y^3 - 18x^4y^2 + 50x^4y + 164x^4 - x^3y^6$
 $- 19x^3y^5 - 47x^3y^4 + 61x^3y^3 + 252x^3y^2 + 102x^3y - 296x^3$
 $+ x^2y^7 + 14x^2y^6 + 89x^2y^5 + 253x^2y^4 + 340x^2y^3 + 438x^2y^2$
 $+ 1044x^2y + 592x^2 - xy^7 - 11xy^6 + 10xy^5 + 159xy^4 + 162xy^3 - 338xy^2$
 $- 568xy - 32x + y^8 + 16y^7 + 78y^6 + 86y^5 - 154y^4 - 340y^3 + 264y^2 - 1000y - 2816$

$i = 34, m = 72$, neighbours 2, 5, 6, 10, 12, 18, 23, 26

W	$-216x^5 - 576x^4 - 648x^3 - 648x^2 + 432x + 576$
A	$-864x^8y^2 - 5040x^8y - 43224x^8 + 20736x^5y + 759012x^5 - 864x^4y^4$ $+3600x^4y^3 + 64200x^4y^2 + 633552x^4y - 3680634x^4 - 15552x^3y^3$ $-72576x^3y^2 - 77760x^3y - 632448x^3 - 20736x^2y^4 - 20736x^2y^3$ $+62208x^2y^2 + 124416x^2y - 964224x^2 - 5184xy^5 + 5184xy^4 + 25920xy^3$ $+752100xy^2 - 7915560xy + 3323334x + 1728y^6 + 23040y^5 + 33312y^4 - 550152y^3$ $+5429562y^2 - 5321502y + 384012$
B	$864x^{11}y^2 + 5040x^{11}y + 43224x^{11} - 864x^{10}y^2 - 5040x^{10}y - 43224x^{10}$ $+864x^9y^3 + 5040x^9y^2 + 43224x^9y - 1728x^8y^3 - 9216x^8y^2$ $-81408x^8y - 741708x^8 + 1728x^7y^4 + 3168x^7y^3 + 8976x^7y^2$ $-562512x^7y + 4142406x^7 + 864x^6y^5 + 1584x^6y^4 + 32568x^6y^3$ $-77712x^6y^2 - 77628x^6y - 3227802x^6 + 864x^5y^5 - 144x^5y^4$ $-26760x^5y^3 - 493776x^5y^2 + 4782546x^5y - 914604x^5 + 864x^4y^6$ $-144x^4y^5 + 31128x^4y^4 - 83520x^4y^3 - 284160x^4y^2 - 828012x^4y$ $-191808x^4 - 1728x^3y^6 - 33408x^3y^5 - 187968x^3y^4 - 827628x^3y^3$ $-699624x^3y^2 + 3930822x^3y - 2438064x^3 + 1728x^2y^7 - 8064x^2y^6$ $-9888x^2y^5 - 494568x^2y^4 + 4547238x^2y^3 + 225564x^2y^2$ $-5035674x^2y + 18197400x^2 - 3456xy^7 + 12672xy^6 + 26400xy^5 + 62268xy^4$ $-3719394xy^3 + 4586322xy^2 - 57948xy - 11923176x + 1728y^8 + 19584y^7 + 21792y^6$ $-599208y^5 + 4818834y^4 - 447180y^3 - 2294880y^2 + 19418184y - 1867824$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 1\}, k_{ij} = 2^2$	
α	1
β	$-x^4 + x + y^2$
γ	x^4
$\{i, j\} = \{0, 6\}, k_{ij} = 3$	
α	1
β	$-x^4 + x^3 - x^2 + 2x + y^2 - 1$
γ	$x^4 - x^3 + x^2 - x + 1$
$\{i, j\} = \{0, 14\}, k_{ij} = 2^2 \cdot 7$	
α	-1
β	$x^4 - 2x^3 + 4x^2 - 9x - y^2 + 16$
γ	$-x^4 + 2x^3 - 4x^2 + 8x - 16$
$\{i, j\} = \{0, 15\}, k_{ij} = 2^2 \cdot 3^2$	
α	1
β	$-x^4 - 2x^3 - 4x^2 - 7x + y^2 - 16$
γ	$x^4 + 2x^3 + 4x^2 + 8x + 16$
$\{i, j\} = \{0, 18\}, k_{ij} = 2^2 \cdot 5$	
α	$-x^4 + 5$
β	$x^7 + x^6 - x^4y^2 + x^4 - 4x^3 + 5x + 5y^2$
γ	$x^7 + x^6 + x^5 + x^4 - 4x^3$
$\{i, j\} = \{0, 23\}, k_{ij} = 3 \cdot 5$	
α	$-x + 4$
β	$-x^4 + 4x^3 - 2x^2 - xy^2 + 8x + 4y^2 - 1$
γ	$x^4 - 4x^3 + x^2 - 4x + 1$
$\{i, j\} = \{0, 26\}, k_{ij} = 3 \cdot 7$	
α	$-x^4 + 4$
β	$x^7 + x^6 - x^5 - x^4y^2 - x^4 - 5x^3 + 9x + 4y^2 + 5$
γ	$x^7 + x^6 - x^4 - 5x^3 + 5x + 5$
$\{i, j\} = \{0, 27\}, k_{ij} = 2^2$	
α	$x - 1$
β	$-x^4 + x^3 - x^2 + xy^2 + x - y^2 - 4$
γ	$-x^4 + x^3 - 2x^2 + 2x - 4$
$\{i, j\} = \{0, 33\}, k_{ij} = 2^2 \cdot 3$	
α	$-x + 1$
β	$-x^4 + x^3 + x^2 - xy^2 - x + y^2 - 4$
γ	$x^4 - x^3 - 2x^2 + 2x + 4$

$$\begin{aligned}
&\{i, j\} = \{1, 8\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^2y^2 - xy^3 + y^4 + y \\
&\gamma \quad -y^4 + 2y^3 - 3y^2 + 5y - 10 \\
&\{i, j\} = \{1, 12\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^2y^2 - xy^3 - y^4 - 2y^3 - y - 2 \\
&\gamma \quad y^4 + y^3 + y + 1 \\
&\{i, j\} = \{1, 24\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^2y^2 + xy^2 - xy - x - y^3 + y^2 + 2y \\
&\gamma \quad y^3 + y^2 - y - 1 \\
&\{i, j\} = \{1, 27\}, k_{ij} = 2^5 \cdot 3^2 \\
&\alpha \quad -7x^5 - 49y + 138 \\
&\beta \quad -7x^9 + 7x^8y - 7x^7y^2 + 7x^6y^2 - 14x^6y + 21x^6 - 7x^5y^3 \\
&\quad + 14x^5y^2 - 28x^5y + 68x^4 - 68x^3y + 68x^2y^2 - 19xy^2 - 32xy + 132x + 19y^3 \\
&\quad - 17y^2 + 6y \\
&\gamma \quad -7x^5y^3 + 7x^5y^2 - 14x^5y + 21x^5 + 68y^3 - 19y^2 - 32y + 132 \\
&\{i, j\} = \{1, 32\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad 3x^5 + 27y + 40 \\
&\beta \quad 3x^9 - 3x^8y + 3x^7y^2 + 3x^6y^2 + 6x^6y + 15x^6 - 3x^5y^3 - 6x^5y^2 \\
&\quad - 12x^5y + 22x^4 - 22x^3y + 22x^2y^2 - 5xy^2 + 26xy + 56x + 5y^3 + y^2 - 16y \\
&\gamma \quad 3x^5y^3 + 3x^5y^2 + 6x^5y + 15x^5 + 22y^3 - 5y^2 + 26y + 56 \\
&\{i, j\} = \{1, 33\}, k_{ij} = 2^4 \\
&\alpha \quad x^5 + y \\
&\beta \quad x^9 - x^8y + x^7y^2 + x^6y^2 - 2x^6y - 3x^6 - x^5y^3 + 2x^5y^2 + 4x^5y \\
&\quad - 2x^4 + 2x^3y - 2x^2y^2 - 3xy^2 + 2xy + 8x + 3y^3 - y^2 - 8y \\
&\gamma \quad x^5y^3 + x^5y^2 - 2x^5y - 3x^5 - 2y^3 - 3y^2 + 2y + 8 \\
&\{i, j\} = \{2, 9\}, k_{ij} = 3 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{2, 10\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 + x + y^2 - y \\
&\gamma \quad -x^4 + 1 \\
&\{i, j\} = \{2, 14\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 + x^3 - x^2 + 2x + y^2 - y + 1 \\
&\gamma \quad x^4 - x^3 + x^2 - x \\
&\{i, j\} = \{2, 16\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad x^4 + x^3 + x^2 + 2x + y^2 - y + 1 \\
&\gamma \quad -x^4 - x^3 - x^2 - x \\
&\{i, j\} = \{2, 17\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad x^3 + y^2 - y + 1 \\
&\gamma \quad -x^3 + x \\
&\{i, j\} = \{2, 18\}, k_{ij} = 2 \cdot 11 \\
&\alpha \quad -2x + 5 \\
&\beta \quad 2x^4 - 3x^3 - 3x^2 - 2xy^2 + 2xy - x + 5y^2 - 5y - 2 \\
&\gamma \quad 2x^4 - 3x^3 - x^2 - 4x - 7 \\
&\{i, j\} = \{2, 19\}, k_{ij} = 3 \\
&\alpha \quad x - 1 \\
&\beta \quad -x^4 + x^3 - x^2 + xy^2 - xy + 2x - y^2 + y - 4 \\
&\gamma \quad -x^4 + x^3 - 2x^2 + 2x - 3 \\
&\{i, j\} = \{2, 21\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad -x^4 + 5 \\
&\beta \quad x^7 - x^6 + x^5 - x^4y^2 + x^4y - 4x^4 - x^3 + x^2 + 3x + 5y^2 - 5y + 8
\end{aligned}$$

γ	$x^7 - x^6 + 2x^5 - 3x^4 - x^3 + x^2 - 2x + 3$
$\{i, j\} = \{2, 24\}, k_{ij} = 3^2$	
α	$-x + 1$
β	$-x^4 + x^3 + x^2 - xy^2 + xy - 2x + y^2 - y - 2$
γ	$x^4 - x^3 - 2x^2 + 2x + 3$
$\{i, j\} = \{2, 25\}, k_{ij} = 2 \cdot 3$	
α	$x^4 + 2$
β	$x^7 - x^6 + x^5 + x^4y^2 - x^4y + 2x^4 + 2x^2 + 2y^2 - 2y + 2$
γ	$-x^7 + x^6 - x^4 - 2x^2 + 2x$
$\{i, j\} = \{2, 28\}, k_{ij} = 3$	
α	1
β	$-x^3 + 2x + y^2 - y + 1$
γ	$-x^3 + x$
$\{i, j\} = \{2, 34\}, k_{ij} = 3 \cdot 11$	
α	$2x^4 - 7$
β	$2x^7 - 2x^6 + 2x^4y^2 - 2x^4y + 8x^4 - 11x^3 + 5x^2 + 10x - 7y^2 + 7y - 34$
γ	$-2x^7 + 2x^6 + 2x^5 - 6x^4 + 11x^3 - 5x^2 - 17x + 27$
$\{i, j\} = \{3, 5\}, k_{ij} = 11 \cdot 13$	
α	32
β	$-16x^4 + 8x^3 - 4x^2 + 34x + 32y^2 + 32y + 15$
γ	$16x^4 - 8x^3 + 4x^2 - 2x + 17$
$\{i, j\} = \{3, 18\}, k_{ij} = 2 \cdot 3$	
α	1
β	$-x^3 - x^2 + x + y^2 + y + 2$
γ	$-x^3 - x^2 + 1$
$\{i, j\} = \{3, 21\}, k_{ij} = 2^2$	
α	1
β	$-x^3 + x^2 + x + y^2 + y$
γ	$-x^3 + x^2 - 1$
$\{i, j\} = \{3, 23\}, k_{ij} = 5$	
α	$x - 1$
β	$x^4 - x^3 + 3x^2 + xy^2 + xy - 2x - y^2 - y + 4$
γ	$-x^4 + x^3 - 2x^2 + 2x - 5$
$\{i, j\} = \{3, 25\}, k_{ij} = 2 \cdot 11$	
α	$-x^4 + 4$
β	$-x^7 + x^6 - 3x^5 - x^4y^2 - x^4y + 2x^4 - 2x^3 + 2x + 4y^2 + 4y + 6$
γ	$x^7 - x^6 + 2x^5 - 3x^4 + 2x^3 + 2x - 2$
$\{i, j\} = \{3, 26\}, k_{ij} = 11$	
α	1
β	$-x^3 - x^2 + 2x + y^2 + y + 4$
γ	$-x^3 - x^2 + x + 3$
$\{i, j\} = \{3, 28\}, k_{ij} = 5 \cdot 13$	
α	$-2x + 1$
β	$2x^4 - x^3 - 8x^2 - 2xy^2 - 2xy + 2x + y^2 + y + 21$
γ	$2x^4 - x^3 - 6x^2 + 3x + 20$
$\{i, j\} = \{3, 30\}, k_{ij} = 1$	
α	-1
β	$x^3 - x^2 - 2x - y^2 - y + 2$
γ	$x^3 - x^2 - x + 3$
$\{i, j\} = \{4, 8\}, k_{ij} = 2^5$	
α	-1
β	$x^4 + x^3y + x^2y^2 + xy^3 + y^4 + y$
γ	$-y^4 + 2y^3 - 5y^2 + 9y - 18$
$\{i, j\} = \{4, 21\}, k_{ij} = 2^7$	
α	$-9x^5 + 162y + 32$
β	$9x^9 + 9x^8y + 9x^7y^2 + 18x^6y^2 - 45x^6y + 81x^6 + 9x^5y^3$ $-27x^5y^2 + 45x^5y + 81x^5 + 4x^4 + 4x^3y + 4x^2y^2 + 170xy^2 - 146xy + 324x$ $+166y^3 - 138y^2 + 146y + 324$
γ	$9x^5y^3 - 18x^5y^2 + 45x^5y - 81x^5 + 4y^3 - 170y^2 + 146y - 324$

$$\begin{aligned}
&\{i, j\} = \{4, 24\}, k_{ij} = 3 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - x^3y - x^2y^2 + xy^2 + xy + x + y^3 + y^2 \\
&\gamma \quad -y^3 - y^2 - y - 1 \\
&\{i, j\} = \{4, 32\}, k_{ij} = 2^6 \cdot 3 \\
&\alpha \quad 5x^5 - 75y - 6 \\
&\beta \quad -5x^9 - 5x^8y - 5x^7y^2 + 5x^6y^2 + 20x^6y + 35x^6 + 5x^5y^3 \\
&\quad + 20x^5y^2 + 30x^5y - 84x^4 - 84x^3y - 84x^2y^2 + 9xy^2 + 96xy + 108x + 9y^3 \\
&\quad + 171y^2 + 114y \\
&\gamma \quad -5x^5y^3 - 5x^5y^2 - 20x^5y - 35x^5 - 84y^3 - 9y^2 - 96y - 108 \\
&\{i, j\} = \{5, 6\}, k_{ij} = 2^4 \cdot 5 \\
&\alpha \quad -16 \\
&\beta \quad -8x^4 + 4x^3y - 2x^2y^2 + xy^3 - y^4 + y^3 + 14y^2 - 44y + 72 \\
&\gamma \quad y^4 - 2y^3 - 28y^2 + 72y - 144 \\
&\{i, j\} = \{5, 10\}, k_{ij} = 2^4 \cdot 7 \cdot 11^2 \\
&\alpha \quad 3888 \\
&\beta \quad 1944x^4 - 972x^3y + 486x^2y^2 - 243xy^3 + 81y^4 + 27y^3 + 1278y^2 + 444y + 1000 \\
&\gamma \quad -81y^4 + 54y^3 + 2556y^2 - 3000y + 2000 \\
&\{i, j\} = \{5, 15\}, k_{ij} = 2^6 \cdot 7 \\
&\alpha \quad 16 \\
&\beta \quad 8x^4 - 4x^3y + 2x^2y^2 - xy^3 + y^4 + 2y^3 - 8y^2 - 16y - 96 \\
&\gamma \quad -y^4 - 4y^3 + 16y^2 + 48y + 192 \\
&\{i, j\} = \{5, 26\}, k_{ij} = 2^4 \cdot 3 \cdot 7 \cdot 11 \\
&\alpha \quad -8x^5 + 60y + 1176 \\
&\beta \quad -4x^9 + 2x^8y - x^7y^2 + 3x^6y^2 - 20x^6 - x^5y^3 - 3x^5y^2 + 6x^5y \\
&\quad + 20x^5 + 552x^4 - 276x^3y + 138x^2y^2 - 534xy^2 + 636xy - 504x + 198y^3 + 246y^2 \\
&\quad + 204y + 504 \\
&\gamma \quad -x^5y^3 + 6x^5y^2 - 40x^5 + 138y^3 - 1068y^2 + 1272y - 1008 \\
&\{i, j\} = \{5, 27\}, k_{ij} = 2^9 \cdot 13 \\
&\alpha \quad -8x^5 + 32y + 1088 \\
&\beta \quad -4x^9 + 2x^8y - x^7y^2 + 2x^6y^2 + 4x^6y - 8x^6 - x^5y^3 - 2x^5y^2 \\
&\quad + 544x^4 - 272x^3y + 136x^2y^2 - 336xy^2 - 256xy - 576x + 168y^3 + 144y^2 + 832y \\
&\gamma \quad -x^5y^3 + 4x^5y^2 + 8x^5y - 16x^5 + 136y^3 - 672y^2 - 512y - 1152 \\
&\{i, j\} = \{5, 28\}, k_{ij} = 2^8 \cdot 5 \cdot 13 \\
&\alpha \quad 24x^5 - 288y - 160 \\
&\beta \quad 12x^9 - 6x^8y + 3x^7y^2 - 6x^6y^2 - 36x^6y + 216x^6 + 3x^5y^3 \\
&\quad + 18x^5y^2 - 96x^5y + 400x^4 - 200x^3y + 100x^2y^2 + 376xy^2 - 1872xy + 4320x \\
&\quad - 188y^3 + 792y^2 - 2240y \\
&\gamma \quad 3x^5y^3 - 12x^5y^2 - 72x^5y + 432x^5 + 100y^3 + 752y^2 - 3744y + 8640 \\
&\{i, j\} = \{5, 29\}, k_{ij} = 2^4 \cdot 3^2 \cdot 11^2 \\
&\alpha \quad -104x^5 + 676y + 2616 \\
&\beta \quad -52x^9 + 26x^8y - 13x^7y^2 + 13x^6y^2 + 156x^6y - 364x^6 - 13x^5y^3 \\
&\quad - 65x^5y^2 + 286x^5y - 364x^5 + 1464x^4 - 732x^3y + 366x^2y^2 - 1718xy^2 \\
&\quad - 388xy - 3480x + 1042y^3 - 1186y^2 + 2660y - 3480 \\
&\gamma \quad -13x^5y^3 + 26x^5y^2 + 312x^5y - 728x^5 + 366y^3 - 3436y^2 - 776y - 6960 \\
&\{i, j\} = \{5, 34\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
&\alpha \quad 4x^5 + 5y^2 + 79y - 56 \\
&\beta \quad 2x^9 + 4x^8y - 2x^7y^2 - 6x^6y^2 + 8x^6 + 2x^5y^3 + 4x^5y^2 - 7x^5y \\
&\quad + 8x^5 - 222x^4 + 856x^3y - 428x^2y^2 - 1332xy^2 - 360xy + 337x + 5y^4 + 487y^3 \\
&\quad + 610y^2 - 1119y + 127 \\
&\gamma \quad 10x^7y - 10x^6y - 2x^5y^3 - 2x^5y^2 + 16x^5 - 10x^3y^2 - 55x^3y - 420x^3 \\
&\quad + 10x^2y^2 + 1545x^2y + 420x^2 - 10xy^3 - 55xy^2 - 1910xy - 408y^3 - 1074y^2 \\
&\quad + 65y + 254 \\
&\{i, j\} = \{6, 8\}, k_{ij} = 5 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y + x^3 - x^2y^2 - 2x^2y - x^2 + xy^3 + 3xy^2 + 3xy + x - y^4 - 4y^3 \\
&\quad - 6y^2 - 5y - 1 \\
&\gamma \quad y^4 + 3y^3 + 3y^2 + 5y - 4 \\
&\{i, j\} = \{6, 10\}, k_{ij} = 2 \cdot 3
\end{aligned}$$

α	2
β	$2x^4 - 2x^3y - 2x^3 + 2x^2y^2 + 4x^2y + 2x^2 - 2xy^3 - 6xy^2 - 6xy - 2x + y^4$ $+ 4y^3 + 7y^2 + 4y + 2$
γ	$-y^4 - 4y^3 - 5y^2 - 6y$
$\{i, j\}$	$\{6, 18\}, k_{ij} = 3 \cdot 17$
α	$4y + 15$
β	$6x^4 - 3x + 4y^3 + 13y^2 + y + 9$
γ	$-4x^3y - 9x^3 + 2x^2y - 3x^2 + 4xy^2 + 11xy - 3x + 2y^2 + 14y - 6$
$\{i, j\}$	$\{6, 19\}, k_{ij} = 3$
α	1
β	$x^4 - x^3y - x^3 + x^2y^2 + 2x^2y + x^2 - xy^2 + x + y^3 + y^2 - 1$
γ	$y^3 + 2y^2 + 3y + 2$
$\{i, j\}$	$\{6, 27\}, k_{ij} = 2^2 \cdot 3^2$
α	$7x^5 + 49y - 290$
β	$7x^9 - 7x^8y - 7x^8 + 7x^7y^2 + 14x^7y + 7x^7 - 7x^6y^2 + 7x^6y$ $+ 7x^5y^3 - 276x^4 + 276x^3y + 276x^3 - 276x^2y^2 - 552x^2y - 276x^2$ $+ 227xy^2 - 66xy - 644x - 227y^3 - 112y^2 + 420y + 644$
γ	$7x^5y^3 + 14x^5y^2 + 28x^5y + 7x^5 - 276y^3 - 601y^2 - 894y - 920$
$\{i, j\}$	$\{6, 28\}, k_{ij} = 2^3 \cdot 3^2$
α	$x^5 - 9y - 10$
β	$x^9 - x^8y - x^8 + x^7y^2 + 2x^7y + x^7 - x^6y^2 - 3x^6y + 4x^6 + x^5y^3$ $+ 4x^5y^2 - 4x^5 + 8x^4 - 8x^3y - 8x^3 + 8x^2y^2 + 16x^2y + 8x^2 + xy^2$ $- 42xy + 32x - y^3 + 32y^2 - 32$
γ	$x^5y^3 + 2x^5y^2 + 5x^5 + 8y^3 + 25y^2 - 18y + 40$
$\{i, j\}$	$\{6, 33\}, k_{ij} = 2 \cdot 3 \cdot 5$
α	$x^5 + 3y + 14$
β	$x^9 - x^8y - x^8 + x^7y^2 + 2x^7y + x^7 + x^6y^2 - x^6y - 4x^6 - x^5y^3$ $+ 6x^5y + 4x^5 + 8x^4 - 8x^3y - 8x^3 + 8x^2y^2 + 16x^2y + 8x^2 + 5xy^2 - 8xy$ $- 14x - 5y^3 + 6y^2 + 36y + 14$
γ	$x^5y^3 + 4x^5y^2 + 2x^5y - 3x^5 + 8y^3 + 29y^2 + 16y - 6$
$\{i, j\}$	$\{6, 34\}, k_{ij} = 5 \cdot 17$
α	$-x^5 + 2y + 13$
β	$-x^9 + x^8y + x^8 - x^7y^2 - 2x^7y - x^7 - 2x^6y^2 - 5x^6y - 5x^6$ $+ x^5y^3 + 2x^5y^2 + x^5y - x^5 + 4x^4 - 4x^3y - 4x^3 + 4x^2y^2 + 8x^2y$ $+ 4x^2 + 6xy^2 + 13xy + 33x - 2y^3 + y^2 - 8y + 4$
γ	$-x^5y^3 - 5x^5y^2 - 8x^5y - 6x^5 + 4y^3 + 18y^2 + 25y + 37$
$\{i, j\}$	$\{7, 16\}, k_{ij} = 7^3$
α	32
β	$32x^4 - 32x^3y + 32x^3 + 32x^2y^2 - 64x^2y + 32x^2 - 32xy^3 + 96xy^2 - 96xy$ $+ 32x + 16y^4 - 40y^3 + 4y^2 + 142y - 151$
γ	$-16y^4 + 88y^3 - 188y^2 + 238y - 183$
$\{i, j\}$	$\{7, 22\}, k_{ij} = 3^3 \cdot 5$
α	$2x^5 - 4y + 25$
β	$2x^9 - 2x^8y + 2x^8 + 2x^7y^2 - 4x^7y + 2x^7 - 4x^6 - 2x^5y^3 + 6x^5y^2$ $+ 2x^5 + 35x^4 - 35x^3y + 35x^3 + 35x^2y^2 - 70x^2y + 35x^2 + 4xy^2 + 14xy - 48x$ $- 39y^3 + 87y^2 - 32y + 35$
γ	$2x^5y^3 - 6x^5y^2 + 6x^5y - 6x^5 + 35y^3 - 101y^2 + 119y - 83$
$\{i, j\}$	$\{7, 23\}, k_{ij} = 5 \cdot 7^2$
α	$4x^5 + 16y + 45$
β	$4x^9 - 4x^8y + 4x^8 + 4x^7y^2 - 8x^7y + 4x^7 + 4x^6y^2 + 4x^6 - 4x^5y^3$ $+ 4x^5y^2 + 4x^5 + 65x^4 - 65x^3y + 65x^3 + 65x^2y^2 - 130x^2y + 65x^2$ $+ 49xy^2 - 12xy - 7x - 49y^3 + 77y^2 + 40y - 7$
γ	$4x^5y^3 - 8x^5y^2 + 12x^5y + 65y^3 - 146y^2 + 183y - 72$
$\{i, j\}$	$\{8, 14\}, k_{ij} = 2^2$
α	-1
β	$x^4 - x - y^2 + 2y - 6$
γ	$-x^4 + 2$
$\{i, j\}$	$\{8, 17\}, k_{ij} = 2^2 \cdot 3$
α	$-x - 2$
β	$-x^4 - 2x^3 + x^2 - xy^2 + 2xy - 2x - 2y^2 + 4y - 10$

γ	$x^4 + 2x^3 - 2x^2 - 4x + 2$
$\{i, j\} = \{8, 20\}, k_{ij} = 3 \cdot 5$	
α	$x - 4$
β	$-x^4 + 4x^3 + xy^2 - 2xy + 4x - 4y^2 + 8y - 15$
γ	$-x^4 + 4x^3 - x^2 + 4x + 1$
$\{i, j\} = \{8, 25\}, k_{ij} = 2^3 \cdot 11$	
α	$x^4 - 9$
β	$x^7 - x^6 + x^4y^2 - 2x^4y + 7x^4 - 12x^3 + 2x^2 + 13x - 9y^2 + 18y - 62$
γ	$-x^7 + x^6 + x^5 - 3x^4 + 12x^3 - 2x^2 - 22x + 26$
$\{i, j\} = \{8, 27\}, k_{ij} = 2^7 \cdot 3$	
α	$7x + 2$
β	$-7x^4 - 2x^3 - 21x^2 + 7xy^2 - 14xy + 22x + 2y^2 - 4y - 90$
γ	$-7x^4 - 2x^3 - 28x^2 - 8x - 98$
$\{i, j\} = \{8, 31\}, k_{ij} = 2^6 \cdot 3 \cdot 11$	
α	$7x^4 - 404$
β	$-7x^7 + 14x^6 - 35x^5 + 7x^4y^2 - 14x^4y + 140x^4 + 110x^3 + 32x^2 - 248x$ $-404y^2 + 808y - 1864$
γ	$-7x^7 + 14x^6 - 42x^5 + 112x^4 + 110x^3 + 32x^2 + 156x - 248$
$\{i, j\} = \{8, 33\}, k_{ij} = 2^4 \cdot 5^2$	
α	$-7x - 2$
β	$-7x^4 - 2x^3 + 21x^2 - 7xy^2 + 14xy - 22x - 2y^2 + 4y - 106$
γ	$7x^4 + 2x^3 - 28x^2 - 8x + 98$
$\{i, j\} = \{9, 12\}, k_{ij} = 3^2 \cdot 5$	
α	32
β	$-16x^4 + 24x^3 - 36x^2 + 86x + 32y^2 + 64y + 15$
γ	$16x^4 - 24x^3 + 36x^2 - 54x + 113$
$\{i, j\} = \{9, 14\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 17$	
α	-1
β	$x^4 - 4x^3 + 16x^2 - 65x - y^2 - 2y + 254$
γ	$-x^4 + 4x^3 - 16x^2 + 64x - 258$
$\{i, j\} = \{9, 15\}, k_{ij} = 2^2 \cdot 3$	
α	1
β	$-x^4 + x + y^2 + 2y + 2$
γ	$x^4 + 2$
$\{i, j\} = \{9, 17\}, k_{ij} = 2^2 \cdot 3$	
α	$-x + 2$
β	$-x^4 + 2x^3 - 3x^2 - xy^2 - 2xy + 2x + 2y^2 + 4y + 2$
γ	$x^4 - 2x^3 + 2x^2 - 4x + 6$
$\{i, j\} = \{9, 19\}, k_{ij} = 3 \cdot 17$	
α	$-x + 4$
β	$x^4 - 4x^3 - 2x^2 - xy^2 - 2xy + 4x + 4y^2 + 8y + 19$
γ	$x^4 - 4x^3 - x^2 + 4x + 3$
$\{i, j\} = \{9, 24\}, k_{ij} = 3^2 \cdot 5$	
α	$-x + 4$
β	$-x^4 + 4x^3 - 2x^2 - xy^2 - 2xy + 4x + 4y^2 + 8y + 13$
γ	$x^4 - 4x^3 + x^2 - 4x + 3$
$\{i, j\} = \{9, 28\}, k_{ij} = 2^4 \cdot 3 \cdot 5$	
α	$-3x + 2$
β	$3x^4 - 2x^3 - 15x^2 - 3xy^2 - 6xy - 2x + 2y^2 + 4y + 62$
γ	$3x^4 - 2x^3 - 12x^2 + 8x + 54$
$\{i, j\} = \{9, 31\}, k_{ij} = 2^2 \cdot 5^2$	
α	$x + 8$
β	$-x^4 - 6x^3 + 15x^2 + xy^2 + 2xy - 4x + 8y^2 + 16y + 34$
γ	$-x^4 - 6x^3 + 14x^2 - 16x + 2$
$\{i, j\} = \{10, 17\}, k_{ij} = 3 \cdot 7$	
α	$x + 4$
β	$-x^5 - x^4y - 5x^4 - x^3y^2 - 6x^3y - 5x^3 - 3x^2y^2 - 5x^2y - 4x^2 + xy^3$ $+ 8xy^2 + 14xy - x + 4y^3 + 16y^2 + 8y - 5$
γ	$-xy^3 - 4xy^2 - 6xy - x - 4y^3 - 16y^2 - 24y - 3$

$$\begin{aligned}
&\{i, j\} = \{10, 20\}, k_{ij} = 3 \cdot 5 \\
&\alpha \quad -2x^5 + 12y + 17 \\
&\beta \quad 2x^9 + 2x^8y + 2x^8 + 2x^7y^2 + 4x^7y + 2x^7 + 2x^6y^2 + 6x^6 + 2x^5y^3 \\
&\quad + 2x^5y^2 + 8x^5y + 6x^5 + x^4 + x^3y + x^3 + x^2y^2 + 2x^2y + x^2 + 13xy^2 \\
&\quad - 6xy + 27x + 13y^3 - 5y^2 + 4y + 27 \\
&\gamma \quad 2x^5y^3 + 4x^5y^2 + 6x^5y - 4x^5 + y^3 - 10y^2 + 9y - 26 \\
&\{i, j\} = \{10, 30\}, k_{ij} = 3 \cdot 5 \cdot 7^2 \\
&\alpha \quad -13x^5 + 338y + 583 \\
&\beta \quad 13x^9 + 13x^8y + 13x^8 + 13x^7y^2 + 26x^7y + 13x^7 + 26x^6y^2 - 13x^6y \\
&\quad + 117x^6 + 13x^5y^3 + 65x^5y + 221x^5 - 206x^4 - 206x^3y - 206x^3 \\
&\quad - 206x^2y^2 - 412x^2y - 206x^2 - 74xy^2 - 431xy + 395x + 132y^3 - 299y^2 - 432y + 996 \\
&\gamma \quad 13x^5y^3 + 13x^5y^2 + 52x^5y - 104x^5 - 206y^3 - 544y^2 - 187y - 601 \\
&\{i, j\} = \{10, 34\}, k_{ij} = 3^4 \\
&\alpha \quad -4x + 7 \\
&\beta \quad 4x^5 + 4x^4y - 3x^4 + 4x^3y^2 + x^3y - 3x^3 - 7x^2y^2 - 10x^2y + 5x^2 \\
&\quad - 4xy^3 - 8xy^2 + 5xy - 17x + 7y^3 + 14y^2 - 21y - 23 \\
&\gamma \quad 4xy^3 + 12xy^2 + 8xy - 8x - 7y^3 - 21y^2 - 14y + 30 \\
&\{i, j\} = \{11, 12\}, k_{ij} = 3^2 \cdot 5 \cdot 7 \cdot 13 \\
&\alpha \quad 1024 \\
&\beta \quad -512x^4 - 256x^3y - 128x^2y^2 - 64xy^3 - 16y^4 - 8y^3 + 516y^2 - 514y + 1 \\
&\gamma \quad 16y^4 - 8y^3 + 516y^2 - 2y + 1 \\
&\{i, j\} = \{11, 16\}, k_{ij} = 2^6 \cdot 3 \cdot 7 \\
&\alpha \quad -16 \\
&\beta \quad 8x^4 + 4x^3y + 2x^2y^2 + xy^3 + y^4 - 2y^3 + 24y^2 - 80y + 352 \\
&\gamma \quad -y^4 + 4y^3 - 48y^2 + 176y - 704 \\
&\{i, j\} = \{11, 17\}, k_{ij} = 2^7 \cdot 3^5 \\
&\alpha \quad 104x^5 - 2704y + 7776 \\
&\beta \quad -52x^9 - 26x^8y - 13x^7y^2 + 26x^6y^2 + 312x^6y + 1352x^6 + 13x^5y^3 \\
&\quad + 156x^5y^2 + 624x^5y - 4096x^4 - 2048x^3y - 1024x^2y^2 - 3360xy^2 - 592xy \\
&\quad + 5408x - 1680y^3 + 1056y^2 - 1184y \\
&\gamma \quad -13x^5y^3 - 52x^5y^2 - 624x^5y - 2704x^5 - 1024y^3 + 6720y^2 + 1184y - 10816 \\
&\{i, j\} = \{11, 25\}, k_{ij} = 2^6 \cdot 3 \cdot 5^2 \\
&\alpha \quad 88x^5 - 484y + 1200 \\
&\beta \quad -44x^9 - 22x^8y - 11x^7y^2 + 11x^6y^2 + 198x^6y + 484x^6 + 11x^5y^3 \\
&\quad + 110x^5y^2 + 396x^5y + 484x^5 - 776x^4 - 388x^3y - 194x^2y^2 - 774xy^2 \\
&\quad + 368xy + 1936x - 290y^3 - 348y^2 + 736y + 1936 \\
&\gamma \quad -11x^5y^3 - 22x^5y^2 - 396x^5y - 968x^5 - 194y^3 + 1548y^2 - 736y - 3872 \\
&\{i, j\} = \{11, 26\}, k_{ij} = 2^4 \cdot 3 \cdot 13 \\
&\alpha \quad 8x^5 + 12y + 72 \\
&\beta \quad -4x^9 - 2x^8y - x^7y^2 - x^6y^2 + 16x^6y - 20x^6 - x^5y^3 + 9x^5y^2 \\
&\quad - 30x^5y + 20x^5 - 72x^4 - 36x^3y - 18x^2y^2 + 6xy^2 + 108xy - 168x - 6y^3 \\
&\quad + 42y^2 - 228y + 168 \\
&\gamma \quad -x^5y^3 + 2x^5y^2 - 32x^5y + 40x^5 - 18y^3 - 12y^2 - 216y + 336 \\
&\{i, j\} = \{12, 26\}, k_{ij} = 2^3 \cdot 3^3 \cdot 11^2 \cdot 13 \\
&\alpha \quad -320x^5 + 6400y + 119656 \\
&\beta \quad -160x^9 + 80x^8y + 80x^8 - 40x^7y^2 - 80x^7y - 40x^7 + 120x^6y^2 + 160x^6y \\
&\quad - 1720x^6 - 40x^5y^3 - 200x^5y^2 + 520x^5y + 2600x^5 + 56148x^4 - 28074x^3y \\
&\quad - 28074x^3 + 14037x^2y^2 + 28074x^2y + 14037x^2 - 54911xy^2 + 37932xy - 45689x \\
&\quad + 20437y^3 + 45545y^2 + 4719y + 61515 \\
&\gamma \quad -40x^5y^3 + 120x^5y^2 + 200x^5y - 3480x^5 + 14037y^3 - 67711y^2 + 117975y \\
&\quad - 77341 \\
&\{i, j\} = \{12, 30\}, k_{ij} = 2 \cdot 3 \cdot 5^2 \\
&\alpha \quad 112x^5 + 532y^2 + 2212y + 1730 \\
&\beta \quad 168x^9 - 56x^8y - 56x^8 + 28x^7y^2 + 56x^7y + 28x^7 - 28x^6y^2 - 448x^6y \\
&\quad + 1932x^6 + 28x^5y^3 + 84x^5y^2 - 1064x^5y + 1064x^5 + 8833x^4 + 1282x^3y \\
&\quad + 1282x^3 - 641x^2y^2 - 1282x^2y - 641x^2 + 6913xy^2 - 10352xy - 12244x + 532y^4 \\
&\quad - 1089y^3 - 12324y^2 - 12982y - 5923 \\
&\gamma \quad 224x^8 + 56x^7y - 168x^7 - 280x^6y - 56x^6 + 28x^5y^3 - 28x^5y^2 - 420x^5y \\
&\quad + 4116x^5 - 1064x^3y^2 - 5908x^3y + 12436x^3 + 1064x^2y^2 + 17305x^2y \\
&\quad - 1039x^2 + 1064xy^3 + 5908xy^2 - 23833xy - 11397x - 2769y^3 - 12374y^2 - 15060y
\end{aligned}$$

$$\begin{aligned}
& \{i, j\} = \{12, 34\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7^2 \\
& \alpha \quad 80x^5 + 400y + 856 \\
& \beta \quad 40x^9 - 20x^8y - 20x^8 + 10x^7y^2 + 20x^7y + 10x^7 + 30x^6y^2 + 40x^6y \\
& \quad + 210x^6 - 10x^5y^3 + 10x^5y^2 - 30x^5y + 110x^5 + 748x^4 - 374x^3y \\
& \quad - 374x^3 + 187x^2y^2 + 374x^2y + 187x^2 - 239xy^2 - 772xy + 7x + 213y^3 + 747y^2 \\
& \quad + 319y + 97 \\
& \gamma \quad 10x^5y^3 + 90x^5y^2 + 110x^5y + 430x^5 + 187y^3 + 83y^2 - 983y + 201 \\
& \{i, j\} = \{13, 15\}, k_{ij} = 2^4 \cdot 3^3 \\
& \alpha \quad 16 \\
& \beta \quad 8x^4 - 4x^3y + 4x^3 + 2x^2y^2 - 4x^2y + 2x^2 - xy^3 + 3xy^2 - 3xy + x + y^4 \\
& \quad - 3y^3 - 11y^2 - 33y - 122 \\
& \gamma \quad -y^4 + 2y^3 + 28y^2 + 78y + 245 \\
& \{i, j\} = \{13, 21\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
& \alpha \quad 8x^5 + 16y + 56 \\
& \beta \quad 4x^9 - 2x^8y + 2x^8 + x^7y^2 - 2x^7y + x^7 - x^6y^2 - 14x^6y + 11x^6 \\
& \quad + x^5y^3 + 4x^5y^2 - 21x^5y + 16x^5 + 52x^4 - 26x^3y + 26x^3 + 13x^2y^2 \\
& \quad - 26x^2y + 13x^2 - 45xy^2 - 70xy + 175x + 29y^3 - 44y^2 - 145y + 256 \\
& \gamma \quad x^5y^3 - 5x^5y^2 - 25x^5y + 21x^5 + 13y^3 - 129y^2 - 101y + 337 \\
& \{i, j\} = \{14, 19\}, k_{ij} = 5^2 \cdot 17 \\
& \alpha \quad -6x^5 + 36y + 523 \\
& \beta \quad -6x^9 + 6x^8y + 12x^8 - 6x^7y^2 - 24x^7y - 24x^7 + 6x^6y^2 + 6x^6y - 18x^6 \\
& \quad - 6x^5y^3 - 18x^5y^2 + 36x^5 + 493x^4 - 493x^3y - 986x^3 + 493x^2y^2 \\
& \quad + 1972x^2y + 1972x^2 - 529xy^2 - 307xy + 759x + 529y^3 + 1401y^2 + 378y - 1518 \\
& \gamma \quad -6x^5y^3 - 30x^5y^2 - 66x^5y - 66x^5 + 493y^3 + 2429y^2 + 5609y + 4703 \\
& \{i, j\} = \{14, 20\}, k_{ij} = 3^2 \\
& \alpha \quad 23x^5 - 90y^2 - 333y - 282 \\
& \beta \quad -2x^9 + 2x^8y + 4x^8 - 2x^7y^2 - 8x^7y - 8x^7 + 2x^6y^2 + 6x^6y + 119x^6 \\
& \quad - 2x^5y^3 + 15x^5y^2 - 58x^5y - 213x^5 + 93x^4 - 93x^3y - 186x^3 \\
& \quad + 93x^2y^2 + 372x^2y + 372x^2 - 129xy^2 - 117xy + 177x - 90y^4 - 60y^3 + 360y^2 \\
& \quad + 348y - 108 \\
& \gamma \quad -25x^8 + 25x^6y + 25x^6 - 2x^5y^3 - 10x^5y^2 - 18x^5y + 103x^5 \\
& \quad + 90x^3y^2 - 81x^3y - 246x^3 - 90xy^3 - 9xy^2 + 327xy + 246x + 93y^3 + 429y^2 \\
& \quad + 999y + 921 \\
& \{i, j\} = \{14, 24\}, k_{ij} = 3^2 \cdot 5^2 \\
& \alpha \quad 4x^5 - 16y + 43 \\
& \beta \quad 4x^9 - 4x^8y - 8x^8 + 4x^7y^2 + 16x^7y + 16x^7 + 4x^6y^2 + 4x^6y - 4x^6 \\
& \quad - 4x^5y^3 - 12x^5y^2 + 8x^5 + 71x^4 - 71x^3y - 142x^3 + 71x^2y^2 \\
& \quad + 284x^2y + 284x^2 + 87xy^2 + 35xy - 103x - 87y^3 - 225y^2 + 76y + 206 \\
& \gamma \quad 4x^5y^3 + 28x^5y^2 + 52x^5y + 28x^5 + 71y^3 + 513y^2 + 887y + 465 \\
& \{i, j\} = \{14, 26\}, k_{ij} = 3 \cdot 7 \cdot 11 \\
& \alpha \quad -4x^5 + 16y + 377 \\
& \beta \quad -4x^9 + 4x^8y + 8x^8 - 4x^7y^2 - 16x^7y - 16x^7 + 8x^6y^2 + 16x^6y + 12x^6 \\
& \quad - 4x^5y^3 - 16x^5y^2 - 16x^5y - 4x^5 + 349x^4 - 349x^3y - 698x^3 \\
& \quad + 349x^2y^2 + 1396x^2y + 1396x^2 - 714xy^2 - 1312xy - 1379x + 365y^3 + 1376y^2 \\
& \quad + 1504y + 1345 \\
& \gamma \quad -4x^5y^3 - 16x^5y^2 - 32x^5y - 20x^5 + 349y^3 + 1380y^2 + 2876y + 1413 \\
& \{i, j\} = \{14, 28\}, k_{ij} = 2^3 \cdot 5^2 \cdot 11 \\
& \alpha \quad 21x^5 - 441y - 1432 \\
& \beta \quad 21x^9 - 21x^8y - 42x^8 + 21x^7y^2 + 84x^7y + 84x^7 - 21x^6y^2 - 84x^6y \\
& \quad + 63x^6 + 21x^5y^3 + 126x^5y^2 + 126x^5y - 126x^5 - 634x^4 + 634x^3y \\
& \quad + 1268x^3 - 634x^2y^2 - 2536x^2y - 2536x^2 + 1075xy^2 + 688xy + 3012x - 1075y^3 \\
& \quad - 3279y^2 - 5820y - 6024 \\
& \gamma \quad 21x^5y^3 + 105x^5y^2 + 168x^5y + 231x^5 - 634y^3 - 2729y^2 - 6920y - 2060 \\
& \{i, j\} = \{15, 16\}, k_{ij} = 2^3 \cdot 3^2 \\
& \alpha \quad 2 \\
& \beta \quad 2x^4 - 2x^3y + 4x^3 + 2x^2y^2 - 8x^2y + 8x^2 - 2xy^3 + 12xy^2 - 24xy + 16x + y^4 \\
& \quad - 6y^3 + 9y^2 + 17y - 46 \\
& \gamma \quad -y^4 + 10y^3 - 39y^2 + 79y - 78 \\
& \{i, j\} = \{15, 17\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7
\end{aligned}$$

α	$x^5 - 7y + 78$
β	$x^9 - x^8y + 2x^8 + x^7y^2 - 4x^7y + 4x^7 + x^6y^2 - 2x^6y - x^6$ $-x^5y^3 + 4x^5y^2 - 2x^5y - 2x^5 + 78x^4 - 78x^3y + 156x^3 + 78x^2y^2$ $-312x^2y + 312x^2 + 85xy^2 - 128xy + 48x - 85y^3 + 291y^2 - 226y + 96$
γ	$x^5y^3 - 5x^5y^2 + 10x^5y - 9x^5 + 78y^3 - 383y^2 + 808y - 576$
$\{i, j\} = \{15, 18\}, k_{ij} = 2^2 \cdot 3$	
α	$x^5 - 6y + 10$
β	$x^9 - x^8y + 2x^8 + x^7y^2 - 4x^7y + 4x^7 - 2x^6y^2 + 9x^6y - 13x^6$ $+x^5y^3 - 5x^5y^2 + 11x^5y - 5x^5 + 4x^4 - 4x^3y + 8x^3 + 4x^2y^2$ $-16x^2y + 16x^2 - 2xy^2 + 36xy - 46x - 2y^3 - 20y^2 + 44y - 14$
γ	$x^5y^3 - 8x^5y^2 + 21x^5y - 21x^5 + 4y^3 - 26y^2 + 84y - 78$
$\{i, j\} = \{15, 23\}, k_{ij} = 3^2$	
α	1
β	$x^4 - x^3y + 2x^3 + x^2y^2 - 4x^2y + 4x^2 + xy^2 - xy - x - y^3 + 3y^2 - 2$
γ	$y^3 - 5y^2 + 11y - 9$
$\{i, j\} = \{15, 26\}, k_{ij} = 3^2 \cdot 11$	
α	$4x^5 - 48y + 45$
β	$4x^9 - 4x^8y + 8x^8 + 4x^7y^2 - 16x^7y + 16x^7 - 8x^6y^2 + 32x^6y - 28x^6$ $+4x^5y^3 - 16x^5y^2 + 16x^5y + 4x^5 + 9x^4 - 9x^3y + 18x^3 + 9x^2y^2$ $-36x^2y + 36x^2 + 30xy^2 + 84xy - 51x - 39y^3 - 48y^2 + 72y + 21$
γ	$4x^5y^3 - 32x^5y^2 + 80x^5y - 60x^5 + 9y^3 - 24y^2 + 192y - 123$
$\{i, j\} = \{15, 27\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7$	
α	$5x^5 - 25y + 92$
β	$5x^9 - 5x^8y + 10x^8 + 5x^7y^2 - 20x^7y + 20x^7 - 5x^6y^2 + 20x^6y - 45x^6$ $+5x^5y^3 - 30x^5y^2 + 90x^5y - 90x^5 + 102x^4 - 102x^3y + 204x^3$ $+102x^2y^2 - 408x^2y + 408x^2 - 77xy^2 + 468xy - 748x + 77y^3 - 647y^2 + 1776y$ -1496
γ	$5x^5y^3 - 35x^5y^2 + 80x^5y - 85x^5 + 102y^3 - 689y^2 + 1692y - 1564$
$\{i, j\} = \{15, 31\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \cdot 11$	
α	$5x^5 - 75y + 222$
β	$5x^9 - 5x^8y + 10x^8 + 5x^7y^2 - 20x^7y + 20x^7 + 5x^6y^2 - 30x^6y - 5x^6$ $+5x^5y^3 - 10x^5y^2 + 10x^5y - 100x^5 + 42x^4 - 42x^3y + 84x^3$ $+42x^2y^2 - 168x^2y + 168x^2 + 117xy^2 - 132xy + 288x - 33y^3 + 21y^2 + 414y + 480$
γ	$5x^5y^3 - 25x^5y^2 + 30x^5y - 45x^5 + 42y^3 - 135y^2 + 372y - 48$
$\{i, j\} = \{15, 32\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5$	
α	$x^5 + 9y + 38$
β	$x^9 - x^8y + 2x^8 + x^7y^2 - 4x^7y + 4x^7 + x^6y^2 - x^6 - x^5y^3$ $+2x^5y^2 + 2x^5y - 2x^5 + 56x^4 - 56x^3y + 112x^3 + 56x^2y^2 - 224x^2y$ $+224x^2 + 47xy^2 - 18xy - 164x - 47y^3 + 121y^2 + 166y - 328$
γ	$x^5y^3 - 5x^5y^2 + 12x^5y - 9x^5 + 56y^3 - 289y^2 + 654y - 612$
$\{i, j\} = \{15, 33\}, k_{ij} = 2^6 \cdot 3^3$	
α	$5x^5 - 75y + 294$
β	$5x^9 - 5x^8y + 10x^8 + 5x^7y^2 - 20x^7y + 20x^7 + 5x^6y^2 - 20x^6y - 5x^6$ $-5x^5y^3 + 30x^5y^2 - 30x^5y - 10x^5 + 204x^4 - 204x^3y + 408x^3$ $+204x^2y^2 - 816x^2y + 816x^2 + 279xy^2 - 606xy + 636x - 279y^3 + 1089y^2 - 1554y$ $+1272$
γ	$5x^5y^3 - 25x^5y^2 + 40x^5y - 45x^5 + 204y^3 - 945y^2 + 1842y - 996$
$\{i, j\} = \{17, 27\}, k_{ij} = 2 \cdot 3 \cdot 7$	
α	$-4x + 10$
β	$4x^4 - 10x^3 + 2x^2y - 4x^2 - 2xy^2 - 7xy + 8x + 5y^2 + 5y + 9$
γ	$2x^2y + 4x^2 + 2xy^2 - 3xy - 8x - 5y^2 - 5y - 1$
$\{i, j\} = \{17, 33\}, k_{ij} = 2$	
α	0
β	-1
γ	1
$\{i, j\} = \{18, 20\}, k_{ij} = 2 \cdot 3$	
α	$-4y - 10$
β	$-2x^9 - 2x^8 - 4x^7 + 2x^6y - 4x^6 + 2x^5y^2 - 4x^5y - 5x^4 - 4x^3y^2$ $-2x^3y + 9x^3 - 4x^2y^2 - 2x^2y + 4x^2 + 4xy^3 - 2xy^2 - 11xy + 25x + 4y^4$ $-2y^3 - 15y^2 - 6y - 21$

$$\begin{aligned}
&\gamma \quad 2x^9 - 2x^5y^2 + 10x^5y - 4x^5 + 5x^4 - 21x - 4y^4 + 10y^3 + 15y^2 - 9y + 46 \\
&\{i, j\} = \{18, 22\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad -2x^4 + 10 \\
&\beta \quad 2x^7 + 2x^6 - x^5y + x^4y^2 - 2x^4y + x^4 - 8x^3 + 4xy - x - 5y^2 + 4y \\
&\gamma \quad -x^5y + 2x^5 - x^4y^2 - 2x^4y + x^4 + 4xy + x + 5y^2 + 4y \\
&\{i, j\} = \{18, 26\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{18, 34\}, k_{ij} = 3^2 \cdot 7 \cdot 11 \cdot 17 \\
&\alpha \quad -1024x^5 + 1280y^2 - 2368y + 5368 \\
&\beta \quad -512x^9 + 1024x^7y - 512x^7 + 1024x^6y - 512x^6 - 256x^5 + 676x^4 \\
&\quad + 1280x^3y^2 + 832x^3y + 336x^3 + 1280x^2y^2 - 520x^2y + 1012x^2 - 1280xy^3 \\
&\quad + 448xy^2 - 856xy - 5786x - 640y^4 - 416y^3 - 1128y^2 - 4542y - 2893 \\
&\gamma \quad 512x^9 - 1024x^5y^2 + 1536x^5y - 256x^5 - 676x^4 + 6798x + 640y^4 - 2144y^3 \\
&\quad + 2096y^2 + 2674y - 9691 \\
&\{i, j\} = \{20, 21\}, k_{ij} = 5 \\
&\alpha \quad -1 \\
&\beta \quad x^6 + 2x^5 - x^3y - x^3 + xy^2 - xy + 5x + y^3 + 3y^2 - 4y + 9 \\
&\gamma \quad -x^6 - x^5 + 3xy - 4x - y^3 - 3y^2 + y - 5 \\
&\{i, j\} = \{21, 25\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad 2x^4 + 6 \\
&\beta \quad -2x^7 + 2x^6 + x^5y - x^4y^2 - 2x^4y + x^4 - 8x^3 + 4xy + x - 3y^2 - 4y \\
&\gamma \quad x^5y - 2x^5 + x^4y^2 - 2x^4y + x^4 + 4xy - x + 3y^2 - 4y \\
&\{i, j\} = \{21, 28\}, k_{ij} = 2^8 \cdot 3 \cdot 5 \\
&\alpha \quad -20 \\
&\beta \quad 2x^9 - 2x^8 + 6x^7 + 2x^6y - 6x^6 - 2x^5y^2 + 6x^5y + 51x^4 + 4x^3y^2 \\
&\quad + 22x^3y - 43x^3 - 4x^2y^2 - 22x^2y + 145x^2 - 4xy^3 - 18xy^2 + 65xy - 750x \\
&\quad + 4y^4 + 18y^3 - 65y^2 + 165y - 1210 \\
&\gamma \quad -2x^9 + 2x^5y^2 - 14x^5y + 6x^5 - 51x^4 + 605x - 4y^4 - 10y^3 + 105y^2 - 375y \\
&\quad + 1960 \\
&\{i, j\} = \{21, 32\}, k_{ij} = 2^8 \\
&\alpha \quad 112x^5 - 280y^2 - 308y + 576 \\
&\beta \quad -14x^9 - 98x^8 + 28x^7y + 126x^7 + 70x^6y - 126x^6 - 70x^5y^2 + 14x^5y \\
&\quad - 65x^4 + 140x^3y^2 + 14x^3y - 707x^3 - 140x^2y^2 + 116x^2y + 837x^2 \\
&\quad - 140xy^3 + 126xy^2 + 591xy - 650x + 140y^4 + 154y^3 - 283y^2 - 113y - 374 \\
&\gamma \quad 14x^9 + 42x^5y^2 - 210x^5y + 126x^5 + 65x^4 - 187x - 140y^4 + 126y^3 + 41y^2 \\
&\quad - 1315y + 1024 \\
&\{i, j\} = \{21, 33\}, k_{ij} = 2^{10} \cdot 3 \cdot 5 \\
&\alpha \quad 112x^5 - 280y^2 + 252y + 4072 \\
&\beta \quad -14x^9 - 98x^8 + 28x^7y + 70x^7 + 70x^6y - 70x^6 - 70x^5y^2 - 42x^5y \\
&\quad - 677x^4 + 140x^3y^2 + 14x^3y - 3983x^3 - 140x^2y^2 + 1340x^2y + 2629x^2 \\
&\quad - 140xy^3 + 126xy^2 + 2643xy - 1598x + 140y^4 + 154y^3 - 2895y^2 - 3505y + 2062 \\
&\gamma \quad 14x^9 + 42x^5y^2 - 98x^5y + 70x^5 + 677x^4 - 1031x - 140y^4 + 126y^3 + 1429y^2 \\
&\quad - 1767y - 464 \\
&\{i, j\} = \{22, 27\}, k_{ij} = 2^7 \cdot 3 \cdot 5 \cdot 13^2 \\
&\alpha \quad -656x^5 - 6888y^2 + 29356y + 33392 \\
&\beta \quad -82x^9 + 574x^8 - 164x^7y + 738x^7 + 410x^6y + 738x^6 - 410x^5y^2 - 1394x^5y \\
&\quad + 20615x^4 + 3444x^3y^2 - 25010x^3y - 8677x^3 + 3444x^2y^2 + 16220x^2y \\
&\quad - 49907x^2 + 3444xy^3 - 21566xy^2 + 7543xy - 4358x - 3444y^4 + 14678y^3 + 21813y^2 \\
&\quad - 7799y + 91098 \\
&\gamma \quad 82x^9 + 246x^5y^2 + 2542x^5y + 738x^5 - 20615x^4 - 45549x + 3444y^4 - 7790y^3 \\
&\quad - 27159y^2 - 34565y - 95456 \\
&\{i, j\} = \{23, 28\}, k_{ij} = 3 \cdot 5 \\
&\alpha \quad 16x + 8 \\
&\beta \quad -16x^4 - 8x^3 - 8x^2y - 4x^2 + 8xy^2 - 16xy + 16x + 4y^2 - 6y + 19 \\
&\gamma \quad -8x^2y - 12x^2 - 8xy^2 + 8xy - 24x - 4y^2 + 6y + 1 \\
&\{i, j\} = \{23, 29\}, k_{ij} = 2^3 \\
&\alpha \quad 488y^2 - 2074y - 882 \\
&\beta \quad -61x^6 + 122x^5y - 61x^5 - 244x^3y^2 + 2013x^3y - 2297x^3 + 6236x^2y
\end{aligned}$$

	$-3118x^2 - 244xy^3 + 1769xy^2 + 2834xy + 403x + 244y^4 - 1037y^3 + 5795y^2 - 6819y - 4357$
γ	$61x^6 - 122x^5y + 3118x^3 - 6236x^2y + 418x - 244y^4 + 1037y^3 + 6677y^2 + 3701y + 821$
$\{i, j\}$	$\{23, 34\}, k_{ij} = 2 \cdot 7$
α	-2
β	$x^6 - x^5 + x^3y - x^3 + xy^2 + 2xy + 7x - y^3 - y^2 - 3y - 7$
γ	$-x^6 + 2x^5 - 2xy - 8x + y^3 + y^2 + 5y + 15$
$\{i, j\}$	$\{25, 26\}, k_{ij} = 3 \cdot 11$
α	8
β	$-4x^4 - 8x^2y - 4x^2 + 8xy + 2x - 10y - 1$
γ	$4x^4 + 2x - 8y^2 - 2y + 1$
$\{i, j\}$	$\{25, 28\}, k_{ij} = 2^7 \cdot 3 \cdot 5$
α	$1296x^5 + 216y^2 - 900y + 760$
β	$-486x^9 - 810x^8 - 972x^7y - 162x^7 + 162x^6y + 162x^6 + 162x^5y^2 - 1134x^5y - 1653x^4 + 108x^3y^2 - 558x^3y - 3967x^3 - 108x^2y^2 - 2748x^2y + 661x^2 + 108xy^3 - 450xy^2 - 1219xy - 254x + 108y^4 - 666y^3 - 319y^2 - 607y + 814$
γ	$486x^9 - 1134x^5y^2 + 810x^5y - 162x^5 + 1653x^4 - 407x - 108y^4 + 450y^3 - 1979y^2 + 2487y - 560$
$\{i, j\}$	$\{25, 29\}, k_{ij} = 3 \cdot 11^2$
α	$-104x^4 + 86$
β	$104x^7 - 104x^6 + 52x^5y - 130x^5 - 52x^4y^2 - 130x^4y - 13x^4 + 252x^3 + 73x^2 + 126xy - 146x + 43y^2 - 68y + 92$
γ	$52x^5y + 234x^5 + 52x^4y^2 - 78x^4y - 91x^4 + 126xy + 73x - 43y^2 - 111y - 19$
$\{i, j\}$	$\{26, 29\}, k_{ij} = 2 \cdot 3 \cdot 11$
α	$-12y + 10$
β	$-x^9 - 2x^8 + 2x^6y + 2x^6 + x^5y^2 + 3x^5y + 2x^5 + 3x^4 - 4x^3y^2 + 6x^3y - 6x^3 - 4x^2y^2 + 6x^2y - 12x^2 + 4xy^3 - 6xy^2 + 6xy + 10x + 2y^4 + 3y^3 - 10y^2 + 11y + 32$
γ	$x^9 - x^5y^2 - x^5y - 3x^4 - 16x - 2y^4 + 5y^3 + 2y^2 + y + 6$
$\{i, j\}$	$\{26, 34\}, k_{ij} = 2 \cdot 3$
α	2
β	$x^4 - 2x^2y - 2xy - x - y - 1$
γ	$-x^4 + x + 2y^2 + y - 1$
$\{i, j\}$	$\{27, 33\}, k_{ij} = 2^4 \cdot 3^2$
α	$-2x + 12$
β	$2x^4 - 12x^3 - x^2y + 3x^2 + xy^2 + 8xy - 14x - 6y^2 - 12y - 22$
γ	$-x^2y + x^2 - xy^2 + 4xy - 10x + 6y^2 + 12y + 26$
$\{i, j\}$	$\{28, 30\}, k_{ij} = 5^2$
α	-1
β	$x^6 + 2x^5 - x^3y - 2x^3 + xy^2 - 2xy + 11x + y^3 + 6y^2 - y + 22$
γ	$-x^6 - x^5 + 6xy - 7x - y^3 - 6y^2 - 5y - 15$
$\{i, j\}$	$\{30, 31\}, k_{ij} = 5^2 \cdot 11$
α	$-2y - 17$
β	$-x^9 - x^8 + 2x^7 + x^6y - x^6 + x^5y^2 - 6x^5y - 3x^5 - 29x^4 + 2x^3y^2 + 9x^3y - 9x^3 - 2x^2y^2 - 9x^2y + 38x^2 - 2xy^3 - 9xy^2 + 9xy + 101x - 2y^4 - 3y^3 + 38y^2 - 113y + 343$
γ	$x^9 - x^5y^2 + 3x^5y + 2x^5 + 29x^4 - 130x + 2y^4 + 7y^3 - 18y^2 + 75y - 222$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 4$ and for $x \leq -4$.

i	k_i	$\varphi_i(x)$	$x \geq 4$	$x \leq -4$
0	-1	$-x^5 - 4$	+	-
1	-1	$-x^5 + x^3 + x^2 - 4$	+	-
2	-1	$-x^5 + x - 3$	+	-
3	-1	$-x^5 - x - 5$	+	-
4	-1	$-x^5 - x^3 - x^2 - 4$	+	-
5	-1	$-x^5 + 8x^3 + 2x^2 - 4$	+	-
6	-1	$-x^5 + x^3 + 4x^2 + 4x - 3$	+	-

i	k_i	$\varphi_i(x)$	$x \geq 4$	$x \leq -4$
7	-1	$-x^5 + x^3 - 2x^2 + 2x - 5$	+	-
8	-1	$-x^5 + 2x + 4$	+	-
9	-1	$-x^5 - 2x - 12$	+	-
10	-1	$-x^5 - x^3 + 2x^2 - 2x - 3$	+	-
11	-1	$-x^5 - 8x^3 - 2x^2 - 4$	+	-
12	-1	$-x^5 + 8x^3 + 14x^2 + 7x - 3$	+	-
13	-1	$-x^5 + 8x^3 - 10x^2 + 5x - 5$	+	-
14	-1	$-x^5 + x^3 + 7x^2 + 14x + 4$	+	-
15	-1	$-x^5 + x^3 - 5x^2 + 10x - 12$	+	-
16	-1	$-x^5 - x^3 + 5x^2 - 10x + 4$	+	-
17	-1	$-x^6 - x^5 - x^3 - 4$	+	+
18	1	$-x^6 + 4x^5 - 3x^4 + x^2 + 4$	+	+
19	1	$-x^6 + x^5 + 3x^4 - x^3 - 3x^2 + x + 5$	+	+
20	1	$-x^6 + x^5 - 3x^4 - x^3 - 3x^2 - x + 3$	+	+
21	1	$-x^6 - 2x^5 - 3x^4 - 2x^3 - x^2 + 4$	+	+
22	-1	$-x^6 + 2x^5 - 3x^4 + x^2 - 4$	+	+
23	-1	$-x^6 - x^5 + 3x^4 - x^3 - 3x^2 + x - 3$	+	+
24	-1	$-x^6 - x^5 - 3x^4 - x^3 - 3x^2 - x - 5$	+	+
25	-1	$-x^6 - 4x^5 - 3x^4 - 2x^3 - x^2 - 4$	+	+
26	1	$-x^6 + 4x^5 - 6x^4 + 6x^3 - 5x^2 + 2x + 3$	+	+
27	1	$-x^6 + x^5 + 6x^4 - x^3 - 12x^2 + 2x + 12$	+	+
28	1	$-x^6 + x^5 - 6x^4 - x^3 - 12x^2 - 2x - 4$	+	+
29	1	$-x^6 - 2x^5 + 4x^3 - x^2 - 2x + 5$	+	+
30	1	$-x^6 - 2x^5 - 6x^4 - 8x^3 - 7x^2 - 4x + 3$	+	+
31	1	$-x^6 - 5x^5 - 12x^4 - 9x^3 - 2x^2 + 4$	+	+
32	-1	$-x^6 - x^5 + 6x^4 - x^3 - 12x^2 + 2x + 4$	+	+
33	-1	$-x^6 - x^5 - 6x^4 - x^3 - 12x^2 - 2x - 12$	+	+
34	-1	$-x^6 - 4x^5 - 6x^4 - 8x^3 - 7x^2 - 4x - 5$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	6	-1	(18, 10), (26, 50), (58, 18), (58, 50)
2	7	-1	(6, 22), (6, 86), (10, 34), (10, 98), (22, 38), (30, 110), (34, 58), (42, 2), (50, 74), (62, 14), (74, 34), (94, 46), (106, 66), (114, 74), (126, 78)
2	8	-1	(2, 154), (14, 30), (14, 158), (18, 234), (38, 118), (50, 138), (54, 70), (66, 218), (70, 22), (78, 94), (78, 222), (82, 42), (86, 230), (98, 122), (102, 182), (118, 134), (130, 26), (142, 30), (142, 158), (146, 106), (166, 246), (178, 10), (182, 198), (194, 90), (198, 150), (206, 94), (206, 222), (210, 170), (214, 102), (226, 250), (230, 54), (246, 6)
2	9	-1	(46, 190), (46, 446), (110, 254), (110, 510), (174, 62), (174, 318), (238, 126), (238, 382), (302, 190), (302, 446), (366, 254), (366, 510), (430, 62), (430, 318), (494, 126), (494, 382)
3	2	1	(1, 5), (3, 5), (4, 2), (7, 8), (8, 4)
3	3	1	(2, 9), (2, 14), (3, 8), (3, 17), (3, 26), (5, 6), (5, 17), (8, 20), (8, 21), (11, 0), (11, 4), (11, 23), (14, 4), (14, 24), (14, 26), (17, 2), (17, 12), (20, 4), (20, 5), (20, 18), (21, 2), (21, 20), (23, 4), (23, 8), (23, 15), (26, 11)
3	4	1	(2, 31), (5, 58), (26, 3), (29, 31), (32, 58), (48, 11), (48, 38), (48, 65), (53, 57), (56, 31), (80, 30)
3	5	1	(59, 139), (140, 139), (221, 139)
5	1	1	(0, 1)
5	2	1	(1, 8), (1, 20), (6, 3), (6, 20), (11, 7), (11, 20), (16, 18), (16, 20), (21, 13), (21, 17), (21, 20)
5	3	1	(1, 47), (6, 27), (11, 73), (16, 12), (26, 72), (31, 52), (36, 48), (41, 37), (51, 97), (56, 77), (61, 23), (66, 62), (76, 122), (81, 102), (86, 123), (91, 87), (101, 22), (106, 2), (111, 98), (116, 112)

p	k	B_p	classes (a, b)
7	1	1	(1, 3), (6, 4)
7	2	1	(1, 36), (5, 21), (5, 31), (5, 46), (8, 1), (12, 7), (12, 45), (12, 46), (15, 15), (19, 10), (19, 42), (19, 46), (22, 29), (26, 28), (26, 46), (29, 43), (33, 14), (33, 38), (33, 46), (36, 8), (40, 0), (40, 3), (40, 46), (43, 22), (47, 17), (47, 35)
7	3	1	(26, 220), (47, 95), (75, 318), (96, 95), (124, 73), (145, 95), (194, 95), (222, 269), (243, 95), (271, 24), (292, 95), (320, 122), (341, 95)
7	4	1	(173, 171), (516, 857), (859, 1543), (1202, 2229), (1545, 514), (1888, 1200), (2231, 1886)
11	1	1	(0, 6), (2, 10), (8, 4), (9, 6)
11	2	1	(4, 43), (4, 86), (5, 78), (7, 1), (13, 6), (13, 17), (13, 28), (13, 39), (13, 50), (13, 72), (13, 83), (13, 94), (13, 105), (13, 116), (15, 9), (15, 21), (15, 91), (16, 45), (18, 56), (26, 53), (26, 69), (26, 120), (27, 12), (29, 111), (37, 97), (37, 98), (38, 100), (40, 45), (48, 20), (48, 76), (49, 67), (51, 100), (59, 54), (59, 64), (60, 34), (62, 34), (70, 32), (70, 108), (71, 1), (73, 89), (81, 10), (81, 31), (82, 89), (84, 23), (92, 58), (92, 75), (92, 109), (93, 56), (95, 78), (103, 36), (103, 87), (103, 119), (104, 23), (106, 12), (114, 42), (114, 65), (115, 111), (117, 67)
11	3	1	(4, 234), (37, 1015), (48, 1114), (59, 245), (70, 1070), (81, 927), (114, 14), (125, 1323), (158, 773), (169, 872), (180, 3), (191, 828), (202, 685), (235, 1103), (246, 1081), (279, 531), (290, 630), (301, 1092), (312, 586), (323, 443), (356, 861), (367, 839), (400, 289), (411, 388), (422, 850), (433, 344), (444, 201), (477, 619), (488, 597), (521, 47), (532, 146), (543, 608), (554, 102), (565, 1290), (598, 377), (609, 355), (642, 1136), (653, 1235), (664, 366), (675, 1191), (686, 1048), (719, 135), (730, 113), (739, 61), (739, 182), (739, 303), (739, 424), (739, 545), (739, 666), (739, 787), (739, 908), (739, 1029), (739, 1150), (739, 1271), (763, 894), (774, 993), (785, 124), (796, 949), (807, 806), (840, 1224), (851, 1202), (884, 652), (895, 751), (906, 1213), (917, 707), (928, 564), (961, 982), (972, 960), (1005, 410), (1016, 509), (1027, 971), (1038, 465), (1049, 322), (1082, 740), (1093, 718), (1126, 168), (1137, 267), (1148, 729), (1159, 223), (1170, 80), (1203, 498), (1214, 476), (1247, 1257), (1258, 25), (1269, 487), (1280, 1312), (1291, 1169), (1324, 256)
13	1	1	(2, 1), (3, 0), (3, 7), (6, 2), (6, 10), (7, 6), (8, 10), (10, 3), (11, 8), (12, 8)
13	2	1	(3, 58), (6, 131), (16, 19), (19, 27), (29, 149), (32, 92), (42, 110), (45, 157), (55, 71), (58, 53), (68, 32), (71, 118), (81, 162), (84, 14), (97, 79), (107, 84), (110, 144), (120, 45), (123, 40), (133, 6), (136, 105), (146, 136), (149, 1), (159, 97), (162, 66)
13	3	1	(94, 799), (263, 292), (432, 1982), (601, 1475), (770, 968), (939, 461), (1108, 2151), (1277, 1644), (1446, 1137), (1615, 630), (1784, 123), (1953, 1813), (2122, 1306)
17	1	1	(0, 4), (3, 4), (3, 9), (4, 5), (5, 4), (5, 15), (6, 10), (6, 14), (13, 0), (13, 15), (14, 3), (14, 4), (14, 10), (15, 10)
17	2	1	(12, 55), (13, 223), (29, 21), (30, 172), (46, 276), (47, 121), (63, 242), (64, 70), (80, 208), (81, 19), (97, 174), (98, 257), (114, 140), (115, 206), (131, 106), (132, 155), (148, 72), (149, 104), (165, 38), (166, 53), (182, 4), (183, 2), (199, 259), (200, 240), (216, 225), (217, 189), (233, 191), (234, 138), (250, 157), (251, 87), (267, 123), (268, 36), (284, 89), (285, 274)